# 2 Diode Applications

## INTRODUCTION TO DIODE

After going through the introduction of the analog circuits in the previous chapter, we are now in a position to talk about diode.

**What is a diode?**

- In the fundamentals of semiconductor physics, we have studied about the semiconductor elements that are found naturally like silicon (Si) and germanium (Ge).
- We also studied that how impure semiconductors are formed by doping some different combination of elements of the periodic table. Which finally leads to the formation of P and N-type semiconductors. Now, the question arises as what is the use of these P-type and N-type semiconductors. Symbolic representation of a PN junction diode is-




**Fig. 2.1 Symbol of Diode**

Left side → P-type, which forms the Anode.

Right side → N-type, which forms the Cathode.

- But the next question arises; what is the use of diode?
- Diode has applications in the electronic circuits, and it is one of the most fundamental building blocks of any electronic circuitry.
- One such an application is the rectification of signal, and the circuit which performs the function of rectification is known as the rectifier. It is constructed by using diodes.
- Before going deep into the study of rectifier and its types, it is important to discuss about diodes first.

**Grey Matter Alert!**

Diode is the most basic element of any electronic circuit.

**Rack Your Brain**

Why do we call a diode as a PN junction and not as a NP junction?

Generally diodes are classified into two types:

**1)** Ideal diode

**2)** Practical diode

### Ideal diode

- A diode is said to be in forward biased condition, if the voltage at anode terminal is greater than the voltage at cathode terminal, and current always flows from anode to cathode.

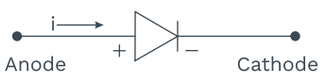



**Fig. 2.2 Diode Under Forward Bias**

- Ideal diode does not require any cut-in voltage, and it is simply represented by a short circuit in forward biased condition.

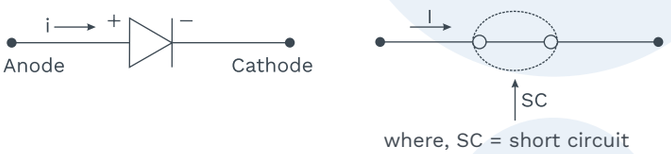



**Fig. 2.3 Ideal Diode Behaviour Under Forward Biased**

### Practical diode

- Usually, all the diodes are practical in nature which drop some cut-in voltage in forward bias to work. A practical diode under forward bias is represented by the voltage source.

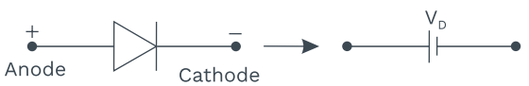



**Fig. 2.4 Practical Diodes Under Forward Biased**

### Load line analysis

As discussed in the fundamentals of semiconductor physics, the current passing through the diode is given by an exponential relation.

$$I_D = I_0\left(e^{\frac{V_D}{\eta V_T}} - 1\right)$$

**Grey Matter Alert!**

AC signal cannot be stored, so it is converted into DC signal using an inverter.

where,

$I_0$ = reverse saturation current
$V_T$ = thermal voltage
$V_T$ = 25 mV at room temperature (300 K)
$V_D$ = voltage across the diode
h = Ideality factor
h = 1 for Si
h = 1.4 for Ge

As the reverse saturation current is very small, so current in the forward bias will be given as.

$$I_D \approx I_0 e^{\frac{V_D}{\eta V_T}} \quad ...(1)$$

The current through a diode in forward biased characteristics is shown below

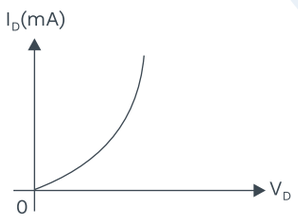



**Fig. 2.5 Practical Diode Characteristics**

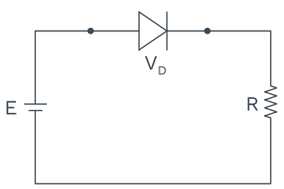



**Fig. 2.6 Series Diode Circuit**

Current through the diode is given by

$$I_D = \frac{E - V_D}{R} \quad ...(2)$$

which is the equation of straight line.

Equations (1) and (2) give load line characteristics of the diode.

**Grey Matter Alert!**

Diode is widely used as switching element.

**Rack Your Brain**

Why does the reverse saturation current play a very important role in the semiconductor?

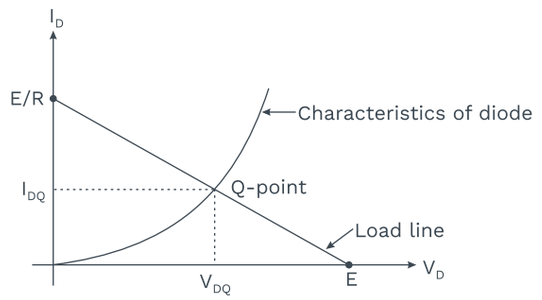



**Fig. 2.7 Load Line Characteristics of Diode**

- After a clear understanding of how a diode works, we are now in a position to study the applications of diode.

**Applications of the diode**

- Rectifier
- Clipper
- Clamper
- Regulator (Zener diode)
- Diode in DC circuits

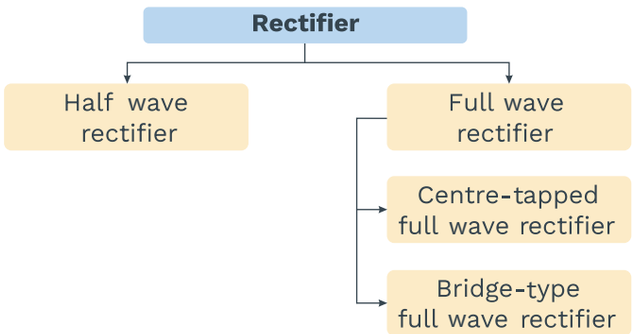
## RECTIFIER

- It is a device, which offers low resistance to the current in one direction and very high resistance to the current in another direction. These devices are very useful in obtaining a constant DC signal from a time varying AC signal.
- Usually, the signal, whose average value over one complete cycle is zero, cannot be stored. So, we require a device, which gives us a signal which can be stored and used whenever required.
- Most of the electronic devices work at DC power supply such as 5 V or 12 V; hence, we have to provide some constant voltage to the devices to work.

**Rack Your Brain**

What is the significance of an operating point?

**Classification of rectifier**




**Half Wave Rectifier**

In the half wave rectifier, only a half of the given wave is rectified, because we use only a single diode for rectification purpose.

**Construction:**

| Components | Purpose |
|---|---|
| Transformer | To step up or step down the input signal |
| Diode | Convert AC to DC signal |

**Circuit diagram**

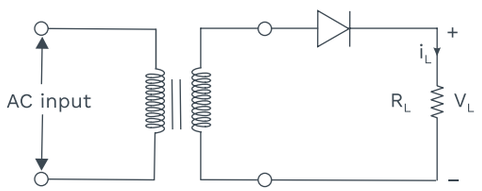



**Fig. 2.8 Circuit Diagram of a Half Wave Rectifier**

**Working**

- To understand the working of a half wave rectifier, consider a sine wave of peak amplitude ($V_m$) and angular frequency (w) such that, $v(t) = V_m \sin(wt)$.

**Grey Matter Alert!**

Rectifier does not convert an AC signal to DC signal but it gives pulsating DC waveform.

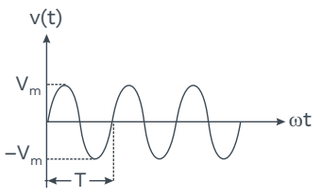



**Fig. 2.9 AC Input Signal**

- When this AC input signal is applied to the input of a step down transformer, a similar output waveform is obtained across the secondary terminal of the transformer, depending on the turns ratio ($N_1$ : $N_2$). For simplification, consider one cycle of input.

**Grey Matter Alert!**

In the rectifier circuit, both step down transformer and step up transformer are used based on the application and the value of input voltage.

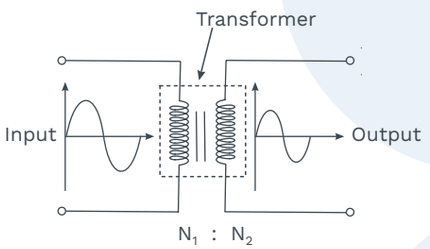



**Fig. 2.10 Working of Transformer**

**Case 1: Positive half cycle**

- During the positive half cycle, the anode terminal of the diode has higher voltage than the cathode terminal, so the diode is in forward biased.

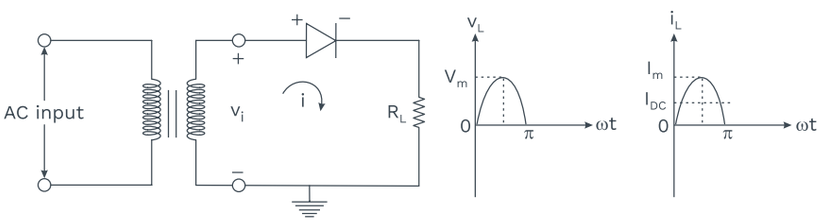



**Fig. 2.11 Working in Positive Half Cycle**

- In the first half cycle, the positive half cycle of the input signal is passed through the diode as the diode is ON.
- The output of the half wave rectifier is taken across load resistance.

- Ideal diode behaves like a short circuit, whereas practical diode is replaced by a cut-in voltage.

**Case 2: Negative half cycle**

- During the negative half cycle, the anode is connected to negative voltage and the cathode is connected to positive voltage, so the diode is in reverse biased i.e. OFF.

**Grey Matter Alert!**

Average value of pure sine wave over a complete cycle is zero.

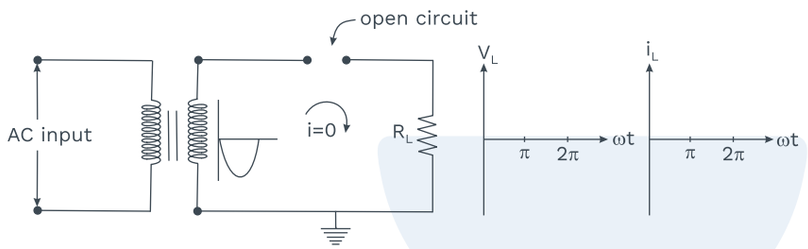



**Fig. 2.12 Working in Negative Half Cycle**

**Waveforms of half wave rectifier**

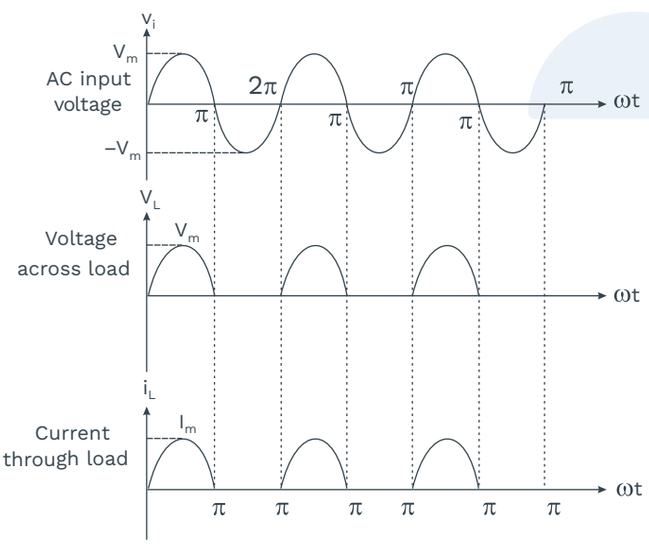



**Fig. 2.13 Input and Output Waveform**

**Parameters of Half wave Rectifier**

**1) DC output current**

The average or DC value of any periodic function is given by

$$I_{DC} = \frac{1}{2\pi}\int_0^{2\pi} i_L \, d(\omega t)$$

$2\pi$ = time period

In half wave rectification

$$I_{DC} = \frac{1}{2\pi}\left[\int_0^{2\pi} I_m \sin(\omega t)\, d(\omega t)\right]$$

$$I_{DC} = \frac{1}{2\pi}\left[\int_0^{\pi} I_m \sin(\omega t)\, d(\omega t) + \int_{\pi}^{2\pi} 0 \, d(\omega t)\right]$$

$$I_{DC} = \frac{I_m}{\pi}$$

**2) RMS output current**

RMS value is also known as the effective value of a periodic function of time, and it is given by

$$I_{rms} = \sqrt{\frac{1}{2\pi}\int_0^{2\pi} i_L^2 d(\omega t)}$$

$$I_{rms} = \left[\frac{1}{2\pi}\int_0^{\pi} I_m^2 \sin^2(\omega t)\, d(\omega t)\right]^{1/2}$$

$$I_{rms} = \frac{I_m}{2}$$

**3) RMS AC output current**

Since output across the load is not pure DC, it has an AC component present as well.

Output current = Output DC current + Output AC current

Output AC current = Output current – Output DC current

$I' = i_L - I_{DC}$

$$I'_{rms} = \sqrt{\frac{1}{2\pi}\int_0^{2\pi} (i_L - I_{DC})^2 \, d(\omega t)}$$

$$I'^2_{rms} = \frac{1}{2\pi}\int_0^{2\pi} (i_L - I_{DC})^2 \, d(\omega t)$$

$$I'^2_{rms} = \frac{1}{2\pi}\left[\int_0^{2\pi} (i_L^2 + I_{DC}^2 - 2I_{DC} \cdot i_L)\, d(\omega t)\right]$$

$$I'^2_{rms} = \frac{1}{2\pi}\int_0^{2\pi} i_L^2 d(\omega t) + I_{DC}^2 \frac{1}{2\pi}\int_0^{2\pi} d(\omega t) - 2I_{DC}\frac{1}{2\pi}\int_0^{2\pi} i_L \, d(\omega t)$$

$$I'^2_{rms} = I^2_{rms} + \frac{I_{DC}^2}{2\pi}(2\pi) - 2I_{DC}(I_{DC})$$

$$I'^2_{rms} = I^2_{rms} + I^2_{DC} - 2I^2_{DC}$$

$$I'^2_{rms} = I^2_{rms} - I^2_{DC}$$

$$I'_{rms} = \sqrt{I^2_{rms} - I^2_{DC}}$$

**4) Ripple factor**

- Rectifier usually converts an AC signal to DC signal, but we have already seen that output current or voltage is not pure DC, but it contains AC component as well, that is AC voltage or current.
- Ripple factor is a numerical value which gives the indication about the amount of AC content present in the final DC signal at the output of the rectifier.
- Low value of ripple factor is desirable.

**Rack Your Brain**

What happens if breakdown occurs in a rectifier circuit?

Ripple factor is given by

$$R = \frac{\text{RMS value of alternating component present in the output}}{\text{Average value of wave}}$$

$$R = \frac{I'_{rms}}{I_{DC}}$$

$$R = \frac{\sqrt{I^2_{rms} - I^2_{DC}}}{I_{DC}} = I_{DC}\frac{\sqrt{\frac{I^2_{rms}}{I^2_{DC}} - 1}}{I_{DC}}$$

$$R = \sqrt{\left(\frac{I_{rms}}{I_{DC}}\right)^2 - 1}$$

Put $I_{rms} = \frac{I_m}{2}$ and $I_{DC} = \frac{I_m}{\pi}$

$$R = \sqrt{\left(\frac{I_m/2}{I_m/\pi}\right)^2 - 1}$$

$$R = \sqrt{\frac{\pi^2}{4} - 1}$$

$R = 1.21$

**Rack Your Brain**

What is the disadvantage of high ripple factor?

RMS AC component is 1.21 times the average component.

Or AC component > DC component, which is undesirable.

**Grey Matter Alert!**

Ripple factor of a rectifier can never be zero.

**5) Input power**

Power is given by voltage × current = $V_i \times i_L$

$$P_i = \frac{1}{2\pi}\int_0^{2\pi} (V_i \times i_L)\, d(\omega t)$$

$V_i$ = voltage across secondary terminal of transformer

$i_L$ = load current

$V_i = i_L R_L$

$$P_i = R_L \frac{1}{2\pi}\int_0^{2\pi} i_L^2\, d(\omega t)$$

$$P_i = I_{rms}^2 R_L$$

In the case of a practical diode, it is represented by its forward resistance $R_F$.

Input power, $P_i > I_{rms}^2 (R_F + R_L)$

**6) Efficiency**

Efficiency is defined as the amount of input converted into output.

In rectifier circuit, Input = input power

Output = output power

Efficiency,

$$\eta = \frac{\text{DC Output Power}}{\text{Input Power}}$$

$$\eta = \frac{P_{DC}}{P_i}$$

$$\eta = \frac{I_{DC}^2 R_L}{I_{rms}^2 (R_F + R_L)}$$

where, $R_F$ = forward resistance of diode,

$R_L$ = load resistance

**Grey Matter Alert!**

Efficiency of any system is always the ratio of same units.

$$\eta = \frac{\frac{I_m^2}{\pi^2} R_L}{\frac{I_m^2}{4}(R_F + R_L)}$$

$$\eta = \frac{4}{\pi^2} \frac{R_L}{(R_F + R_L)}$$

$$\% \ \eta = \eta \times 100$$

$$\% \ \eta = \frac{400}{\pi^2} \frac{R_L}{(R_F + R_L)}$$

$$\% \ \eta = 40.5 \frac{R_L}{R_F + R_L}$$

$\% \ \eta = 40.5$ (if $R_F << R_L$)

**7) Regulation**

Regulation of a half wave rectifier is a very important parameter, and it gives how DC output voltage varies as a function of DC load current.

$$\text{Regulation} = \frac{V_{NL} - V_{FL}}{V_{FL}}$$

$$\% \text{ regulation} = \frac{V_{NL} - V_{FL}}{V_{FL}} \times 100\%$$

$V_{NL}$ = no load voltage
$V_{FL}$ = full load voltage

$V_{NL}$ : it is the output voltage under no load condition (no current drawn from the supply).

$$V_{NL} = \frac{V_{DC}}{I_{DC}}$$

$$V_{NL} = \frac{V_m}{\pi}$$

$V_{FL}$ : it is the output voltage when DC current is not zero.

$V_{FL} = V_{NL}$ – DC voltage drop

**Grey Matter Alert!**

Form factor is fixed for a particular type of waveform.

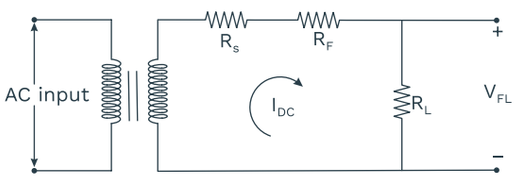



**Fig. 2.14 Half Wave Rectifier**

$R_s$ = secondary resistance of transformer
$R_F$ = forward resistance of the diode.

$$V_{FL} = \frac{V_m}{\pi} - I_{DC}(R_s + R_F)$$

$$V_{FL} = \frac{V_m}{\pi} - I_{DC}R$$

$$\% \text{ Regulation} = \frac{V_{NL} - V_{FL}}{V_{FL}} \times 100$$

$$\% \text{ Regulation} = \frac{\frac{V_m}{\pi} - \left(\frac{V_m}{\pi} - I_{DC}R\right)}{I_{DC}R_L}$$

$$\% \text{ Regulation} = \frac{R}{R_L} \times 100$$

Where, $R = R_F + R_S$

**8) Peak Inverse Voltage**

- During the negative half cycle, the diode is reverse biased, and it is represented by an open circuit.
- Whenever reverse biased voltage is increased, there is a possibility of breakdown of the diode, that is Zener or avalanche breakdown.
- Peak inverse voltage gives maximum reverse biased voltage that a diode can withstand before breakdown.

Input voltage signal = $V_m \sin(wt)$

Maximum input voltage = $|V_m \sin(wt)|$

$|\sin(wt)| = 1$

$\therefore$ maximum input voltage = $V_m$

Peak inverse voltage, PIV = $V_m$

**Previous Years' Questions**

For full wave rectification, a four-diode bridge rectifier is claimed to have the following advantages over a two-diode circuit.

**1)** Less expensive transformer
**2)** Smaller size transformer and
**3)** Suitability for higher voltage application of these

**a)** only 1 and 2 are true
**b)** only 1 and 3 are true
**c)** only 2 and 3 are true
**d)** 1, 2, as well as 3 are true

**9) Transformer Utilisation Factor (TUF)**

It is defined as,

$$TUF = \frac{\text{DC power delivered to the load}}{\text{AC power rating of the transformer}}$$

$$TUF = \frac{2\sqrt{2}}{\pi^2}\frac{R_L}{R_F + R_L}$$

$$TUF = 0.286\frac{R_L}{R_F + R_L}$$

For ideal diode $R_F = 0$

Maximum TUF = 0.286

**10) Angle of conduction**

- In the rectifier circuit, a diode is not always in conducting mode.
- It conducts only for a half cycle in the case of a half wave rectifier.
- Due to the non-ideal nature of the diode, the conductivity time is less.
- The conductivity time will likely get reduced, because the diode takes some time to conduct. When the input voltage is greater than cut-in voltage of the diode, it starts to conduct.

Suppose $V_g$ is the cut-in voltage.

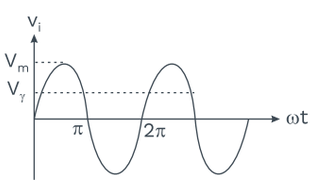



**Fig. 2.15 Input Wave**

Suppose, input waveform reaches $V_g$ when phase of input signal = $\theta$.

That is at $wt = \theta$

$V_m \sin\theta = V_g$

$$\theta = \sin^{-1}\frac{V_\gamma}{V_m}$$

Total conduction angle = $180° - 2\theta$

**Grey Matter Alert!**

Angle of conduction represents the time for which the diode is in 'ON' condition.

**11) Form Factor**

It is the ratio of RMS value to average value.

$$FF = \frac{I_{rms}}{I_{avg}} = \frac{\pi}{2}$$

For half wave rectifier, FF = 1.57

**12) Peak Factor**

It is the ratio of peak value to RMS value.

$$\text{Peak factor (PF)} = \frac{\text{Peak value}}{\text{RMS value}}$$

In half waver rectifier,

$$PF = \frac{V_m}{V_m / 2} = 2$$

**Example 1:**

Find the average and rms value of the given voltage signal x(t) = 10 sin wt, if the wave is passed through a half waver rectifier.

**Sol:**

Given, input signal, x(t) = 10 sin wt

In half wave rectifier output, DC and rms voltage are given by-

$$V_{DC} = \frac{V_m}{\pi} \text{ and } V_{rms} = \frac{V_m}{2}$$

$$V_{DC} = \frac{10}{\pi} = 3.18 \text{ V}$$

$$V_{rms} = \frac{V_m}{2}$$

$$V_{rms} = \frac{10}{2} = 5 \text{ V}$$

**Example 2:**

A voltage signal 2 sin wt is applied to the half rectifier. Find the following

a) RMS, b) Average value, c) Form factor, d) Peak factor, e) Conduction angle, if $V_g$ = 0.7 V and f) Peak inverse voltage.

**Previous Years' Questions**

In a full-wave rectifier using ideal diodes, $V_{dc}$ and $V_m$ are dc and peak values of the voltage, respectively ,across a resistive load, if PIV is the peak inverse voltage of the diode, then the appropriate relationships for this rectifier are

**a)** $V_{ac} = \frac{V_m}{\pi}$, $PIV = 2V_m$

**b)** $V_{dc} = 2\frac{V_m}{\pi}$, $PIV = 2V_m$

**c)** $V_{dc} = 2\frac{V_m}{\pi}$, $PIV = V_m$

**d)** $V_{ac} = \frac{V_m}{\pi}$, $PIV = V_m$

**Sol:**

**a)** RMS value $= \frac{V_m}{2}$

$= \frac{2}{2} = 1V$

**b)** Average value

$= \frac{V_m}{\pi} = \frac{2}{\pi} = 0.6366\,V$

**c)** Form factor = 1.57

**d)** Peak factor = 2

**e)** Conduction angle

$\theta = \sin^{-1} \frac{V_\gamma}{V_m}$

$\theta = \sin^{-1} \frac{0.7}{2}$

$\theta = 20.48°$

Conduction angle = $180° - 2\theta$

$= 180° - 40.97° = 139.03°$

**f)** Peak inverse voltage = $V_m = 2V$

**Centre – Tapped Full Wave Rectifier**

- After going through half wave rectifier, where a single diode is use for rectification, the maximum efficiency is 40.5%, which is less than 50 percent.
- The question is how to increase the efficiency, and the answer is a full wave rectifier.
- In half wave rectifier, a diode conducts only during one half cycle and remains OFF during another half cycle, so the load current flows only for one half cycle.
- If the diode conducts in both the cycles, the efficiency will be doubled.

**Rack Your Brain**

Why should the direction of current flowing through the load be same in both the cycles?

**Grey Matter Alert!**

Centre tap transformer is used in rectifier so that current will find a closed path to flow in both the cycles.

**Construction:**

| Components | Purpose |
|---|---|
| Transformer | To step up or step down input. |
| Diode $D_1$ | Conducts during positive half cycle. |
| Diode $D_2$ | Conducts during negative half cycle. |
| Load resistance | For output voltage. |

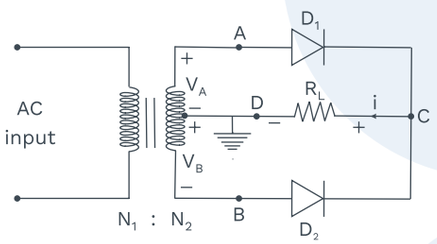



**Fig. 2.16 Full Wave Rectifier**

**Working**

- To understand the working of a full wave rectifier circuit, consider the sine wave.

  $v(t) = V_m \sin \omega t$

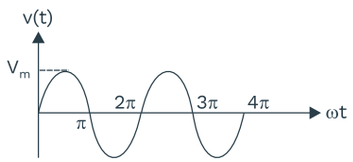



**Fig. 2.17 Input Sine Wave**

- Now this input sine wave is applied to the primary terminals of step up or step down transformer. The secondary turns are divided into two equal number of turns using a centre tap, that's why a full wave rectifier is called a centre tapped full wave rectifier.

**Grey Matter Alert!**

Efficiency of full wave rectifier is double to that of a half wave rectifier.

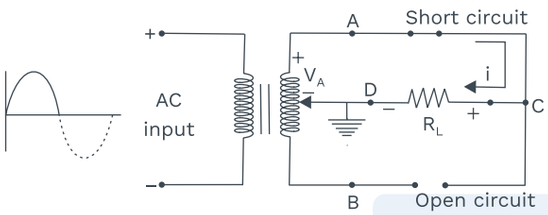
**Rack Your Brain**

What happens if we do not use centre tapped rectifier in a full wave rectifier circuit?

**Case 1: Positive Cycle**

- During positive half cycle, diode $D_1$ is in forward biased and $D_2$ is in reverse biased.
- Diode $D_1$ is represented by short circuit. Diode $D_2$ is represented by open circuit.

Equivalent circuit is given below:



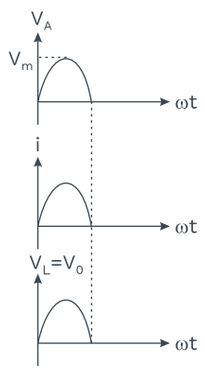




**Fig. 2.18 Output Waveforms**

Applying KVL in secondary loop,

$$V_A - iR_L = 0$$
$$V_A = iR_L$$
$$V_0 = iR_L$$

**Case 2: Negative cycle**

- During the negative half cycle, diode $D_2$ is forward biased and $D_1$ is reverse biased.
- Diode $D_1$ is represented by open circuit. Diode $D_2$ is represented by short circuit.

**Grey Matter Alert!**

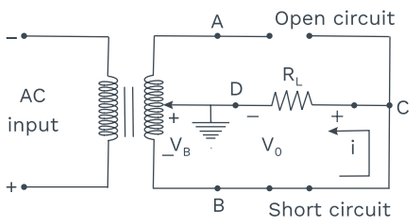
Current flowing through the load in both the cycles should be in the same direction.

**Previous Years' Questions**

A dc power supply has a no-load voltage of 30V and a full-load voltage of 25V at a full-load current of 1A. Its output resistance and load regulation, respectively, are

**a)** 5 Ω and 20%
**b)** 25 Ω and 20%
**c)** 5 Ω and 16.7%
**d)** 25 Ω and 16.7%

Equivalent circuit is shown below:



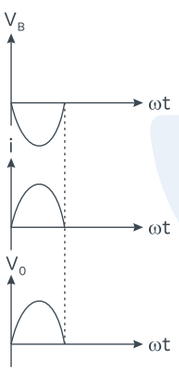




**Fig. 2.19 Output Waveforms**

Applying KVL in the above loop,

$V_B + iR_L = 0$

$V_B = -iR_L$

$V_0 = iR_L = -V_B$

- It is clearly seen that during both the cycles of the input signal, the direction of current is same as the direction of voltage (polarity) in both the cycles.

**Comparison with half wave rectifier**

In half wave rectifier, only half cycle of input is passed, and in full wave rectifier both the cycles of input are passed due to the presence of two diodes.

**Rack Your Brain**

What is the significance of voltage regulation in FWR?

### Waveforms of full wave rectifier

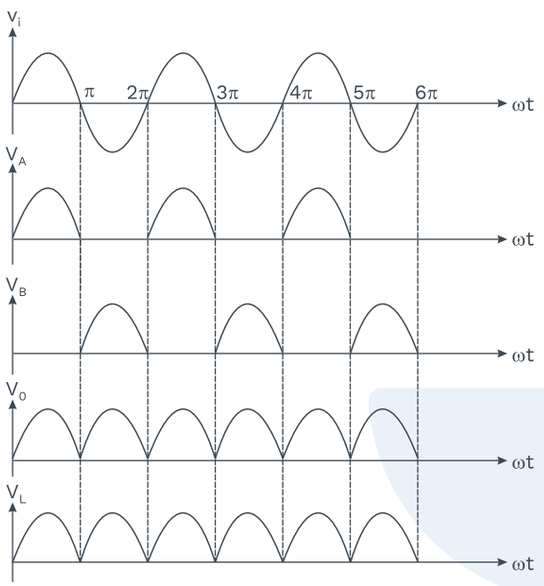



**Fig. 2.20 Output Waveforms**

### Parameters of Full wave Rectifier

**1) DC Output Current**

$$I_{DC} = \frac{1}{2\pi}\int_0^{2\pi} i\, d(\omega t)$$

$$i = I_m \sin(wt)$$

$$I_{DC} = \frac{1}{2\pi}\left[\int_0^{2\pi} I_m \sin(\omega t)\, d(\omega t)\right]$$

$$I_{DC} = \frac{2I_m}{\pi}$$

**2) RMS Output Current**

RMS or effective value is given by

$$I_{rms} = \sqrt{\frac{1}{2\pi}\int_0^{2\pi} i^2 d(\omega t)}$$

$$I_{rms} = \sqrt{\frac{1}{2\pi}\int_0^{2\pi} I_m^2 \sin^2(\omega t)\, d(\omega t)}$$

$$I_{rms} = \frac{I_m}{\sqrt{2}}$$

**Grey Matter Alert!**

Transformer utilisation factor and efficiency are nearly same thing.

**3) Input Power**

Power is given by = voltage × current

During positive half cycle,

Instantaneous power = $V_A \times i$

$$\text{Average power, } P_A = \frac{1}{\pi}\int_0^{\pi} V_A \cdot i\, d(\omega t)$$

During the negative half cycle,

Instantaneous power = $V_B \times i$

$$\text{Average power, } P_B = \frac{1}{\pi}\int_{\pi}^{2\pi} V_B \cdot i\, d(\omega t)$$

Total average power, $P = P_A + P_B$

$$P = \frac{1}{2\pi}\left[\int_0^{\pi} V_A \cdot i\, d(\omega t) + \int_{\pi}^{2\pi} V_B \cdot i\, d(\omega t)\right]$$

$$P = \frac{1}{2\pi}\left[\int_0^{\pi} i \cdot R_L \cdot i\, d(\omega t) + \int_{\pi}^{2\pi} i \cdot R_L \cdot i\, d(\omega t)\right]$$

$$P = R_L \frac{1}{2\pi}\int_{\pi}^{2\pi} i^2 d(\omega t)$$

$$P = I_{rms}^2 R_L$$

**4) RMS Value of AC Output Current**

We have seen in the half wave rectifier, the output of full wave rectifier also contains AC + DC voltage.

R.M.S value of AC output current is given by-

$$I'_{rms} = \sqrt{I_{rms}^2 - I_{DC}^2}$$

**5) Ripple Factor**

Ripple factor is a numerical value, which gives us an indication about the amount of AC component present in the output DC component. It is given by-

$$R = \frac{I'_{rms}}{I_{DC}} = \frac{\sqrt{I_{rms}^2 - I_{DC}^2}}{I_{DC}} = \sqrt{\left(\frac{I_{rms}}{I_{DC}}\right)^2 - 1}$$

$$R = \sqrt{\frac{(I_m/\sqrt{2})^2}{(2I_m/\pi)^2} - 1}$$

**Rack Your Brain**

Which type of rectifier is used in mobile charger?

$$R = \sqrt{\frac{\pi^2}{8} - 1}$$

$$R = 0.483$$

**6) Efficiency**

Since input power is AC and output power is DC, then

$$\text{efficiency, } \eta = \frac{P_{DC}}{P_{AC}}$$

$$\%\eta = \frac{P_{DC}}{P_{AC}} \times 100\%$$

$$\eta = \frac{I_{DC}^2 R_L}{I_{rms}^2 R_L} \times 100\%$$

$$\text{Put } I_{DC} = \frac{2I_m}{\pi}$$

$$I_{rms} = \frac{I_m}{\sqrt{2}}$$

$$\%\,\eta = \frac{800}{\pi^2}\%$$

$$\%\ \eta = 80.05\%$$

**7) Voltage Regulation**

$$\%\ \text{regulation} = \left(\frac{V_{NL} - V_{FL}}{V_{FL}}\right) \times 100$$

**$V_{NL}$:** It is the output voltage under no load condition (no current drawn from the supply).

$$V_{NL} = \frac{V_{DC}}{I_{DC}}$$

It is the average of the output of full wave rectifier, $V_{NL} = \frac{2V_m}{\pi}$

**$V_{FL}$:** It is the output voltage when load is not open circuited, and DC current is not zero.

$$V_{FL} = \frac{V_{DC}}{I_{DC}} \neq 0$$

**Grey Matter Alert!**

RMS current of full wave rectifier is more because it conducts in both the cycles.

$V_{FL} = V_{NL} - I_{DC} R$

R = resistance across the secondary terminal.

$$\text{Regulation} = \left(\frac{V_{NL} - V_{FL}}{V_{FL}}\right) \times 100$$

$$= \frac{\frac{2V_m}{\pi} - \left(\frac{2V_m}{\pi} - I_{DC}R\right)}{I_{DC}R_L}$$

$$\% \text{ Regulation} = \frac{R}{R_L} \times 100\,\%$$

Where, $R = R_F + R_S$

**8) Peak Inverse Voltage (PIV)**

It is the maximum reverse biased voltage that the diode can withstand without breakdown in reverse biased condition.

$PIV = 2V_m$

**9) Transformer Utilisation Factor (TUF)**

$$(TUF) = \frac{\text{DC output power}}{\text{AC rating of primary winding}}$$

$$(TUF) = \frac{I_{DC}^2 R_L}{V_{rms}\, I_{rms}}$$

$$TUF = \frac{I_{DC}^2 R_L}{\frac{V_m}{\sqrt{2}} \times \frac{I_m}{\sqrt{2}}}$$

$$TUF = \frac{\left(\frac{2I_m}{\pi}\right)^2 R_L}{\frac{V_m I_m}{2}}$$

$$TUF = \frac{8}{\pi^2} = 0.81$$

If we consider the diode as a practical diode with forward resistance $R_F$,

$$TUF = 0.81 \frac{R_L}{R_F + R_L}$$

$TUF_{max} = 0.81$ (when $R_F = 0$)

**10) Peak Factor (PF)**

$$\text{Peak Factor, PF} = \frac{I_m}{I_{rms}}$$

$$PF = \frac{I_m}{I_m / \sqrt{2}} = \sqrt{2} = 1.41$$

Peak factor = 1.41

**11) Form Factor (FF)**

$$\text{Form factor, FF} = \frac{I_{rms}}{I_{DC}}$$

$$FF = \frac{I_m / \sqrt{2}}{2I_m / \pi} = \frac{\pi}{2\sqrt{2}}$$

FF = 1.11

**12) Angle of conduction**

- If the diodes are ideal then, the conduction angle should be 360°.
- Due to non-ideal nature of diode, the net conduction angle is slightly less.

Consider a diode has a cut-in voltage of $V_g$.

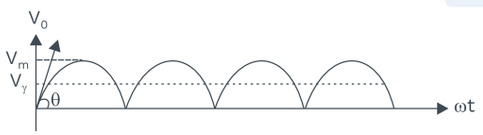



**Fig. 2.21 Output Waveform**

Net angle conduction in one half cycle = 180° – 2θ

Net angle conduction two half cycle = 360° – 4θ

Where, $\theta = \sin^{-1}\frac{V_\gamma}{V_m}$

**Note:**

- Full wave rectifier is better and efficient in almost all the parameters except higher peak inverse voltage.
- Efficiency, DC output voltage, transformer utilisation factor, etc. are almost double as compared to half wave rectifier.

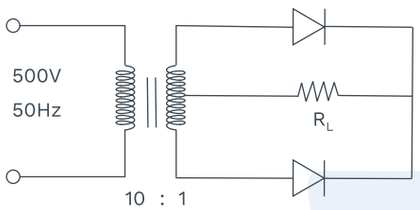
**Grey Matter Alert!**

The full wave rectifier and bridge rectifier are almost same in all performance parameters.

| Parameters | Half wave rectifier | Full wave rectifier |
|---|---|---|
| DC output current | $\frac{I_m}{\pi}$ | $\frac{2I_m}{\pi}$ |
| RMS output current | $\frac{I_m}{2}$ | $\frac{I_m}{\sqrt{2}}$ |
| Ripple factor | 1.21 | 0.483 |
| Efficiency (max) | 40.5 | 80.05 |
| Peak inverse voltage | $V_m$ | $2V_m$ |
| Voltage regulation (%) | $\frac{R}{R_L} \times 100$ | $\frac{R}{R_L} \times 100$ |
| Transformer utilisation factor (max) | 0.286 | 0.81 |
| Output frequency | f | 2f |
| Angle of conduction | $180° - 2q$ | $360° - 4q$ |
| DC diode voltage | $\frac{I_m R_L}{\pi}$ | $\frac{2I_m R_L}{\pi}$ |
| Circuit complexity | Less | More |
| DC output voltage | $\frac{V_m}{\pi}$ | $\frac{2V_m}{\pi}$ |
| RMS output voltage | $\frac{V_m}{2}$ | $\frac{V_m}{\sqrt{2}}$ |

**Table 2.1 Comparison Between Half Wave and Full Waver Rectifier**

**Example 3:**
A full wave rectifier is shown below, which takes output across a load resistance of 800Ω. Two practical diodes are used having an internal resistance of 60Ω each. The transformer turn ratio is 10:1, and R.M.S voltage applied across the primary winding is 500V & 50Hz supply.



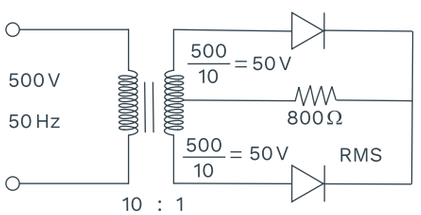

Calculate

**i)** RMS Voltage and current
**ii)** DC Voltage and current
**iii)** PIV (Peak Inverse Voltage)
**iv)** Output frequency
**v)** Ripple factor
**vi)** Voltage regulation
**vii)** Efficiency

**Grey Matter Alert!**

The main disadvantage of bridge rectifier is voltage drop across diodes, which reduces the output voltage.

**Sol:**
The given circuit can be redrawn as shown below.




$R_L$ = 800 Ω

$R_F$ = 60 Ω

**i)** RMS Voltage and current are given by-

$V_m$ = 50

$$V_{rms} = \frac{V_m}{\sqrt{2}} = \frac{50}{\sqrt{2}}$$

$V_{rms}$ = 35.35

$$I_m = \frac{V_m}{R_F + R_L}$$

$$I_m = \frac{50}{60 + 800} = 0.05813\ A$$

$I_m$ = 58.13 mA

$$\text{RMS output current} = \frac{I_m}{\sqrt{2}}$$

$$I_{rms} = \frac{58.13}{\sqrt{2}}$$

$I_{rms}$ = 41.1 mA

**ii)** DC Voltage and current are given by-

$$\text{DC output current} = \frac{2I_m}{\pi}$$

$$I_{DC} = \frac{2 \times 58.13}{\pi} mA$$

$I_{DC}$ = 37 mA

$$V_{DC} = \frac{2V_m}{\pi}$$

$$I_{DC} = \frac{2 \times 50}{\pi}$$

$V_{DC}$ = 31.83 V

**iii)** PIV

Peak inverse voltage = $2V_m$

PIV = 2 × 50

PIV = 100 V

**iv)** Ripple factor

R = 0.483

**v)** Output frequency

$f_0 = 2f_i$

$f_i$ = 50 Hz

$f_0$ = 100 Hz

**vi)** Voltage regulation

$$\% \text{ regulation} = \frac{R_F}{R_L} \times 100$$

$$\% \text{ regulation} \approx 75\%$$

**vii)** Efficiency

$$\% \text{ Efficiency} = 81 \times \frac{800}{R_F + R_L} \%$$

$$= 81 \times \frac{800}{860}$$

$$\% \text{ Efficiency} = 75.34\%$$

**Example 4:**

In the above example, if practical diode has cut-in voltage of 0.6V. Find conduction angle.

**Sol:**

Conduction angle = $360° - 4\theta$

$$\theta = \sin^{-1}\left(\frac{V_\gamma}{V_m}\right)$$

$$\theta = \sin^{-1}\frac{0.6}{50}$$

$$4\theta = 4 \times \sin^{-1}\frac{0.6}{50}$$

$$4\theta = 2.75°$$

Angle of conduction = $360° - 4\theta$

$= 360° - 2.75°$

Angle of conduction = 357.25°

**Bridge Rectifier**

- After going through half wave and full wave rectifiers in detail, the next type of rectifier circuit is a bridge type rectifier.

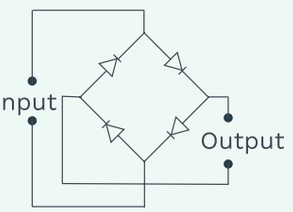
**Why bridge rectifier?**

- Full wave rectifier provides good efficiency, high output voltage and current, low ripple factor and high angle of conduction.

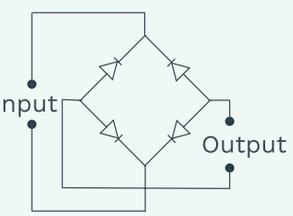
**Previous Years' Questions**

The correct full wave rectifier circuit is

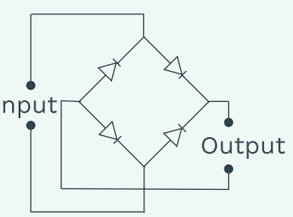
**a)**



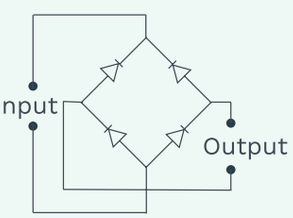

**b)**




**c)**




**d)**



- But the main drawback in the full wave rectifier is centre-tapped transformer, which is the most critical element of the circuit.
- Centre-tapped transformer makes the circuit large and complex and makes the circuit more expensive.
- Another drawback of full wave rectifier is its high peak inverse voltage which may destroy the whole circuit. To overcome these drawbacks, bridge rectifier is used.

**Construction details:**

| Components | Purpose |
|---|---|
| Simple transformer | For step up or step down. |
| Diode $D_1$, $D_3$ | For conduction in positive half cycle. |
| Diode $D_2$, $D_4$ | For conduction in negative half cycle. |

**Table 2.2**

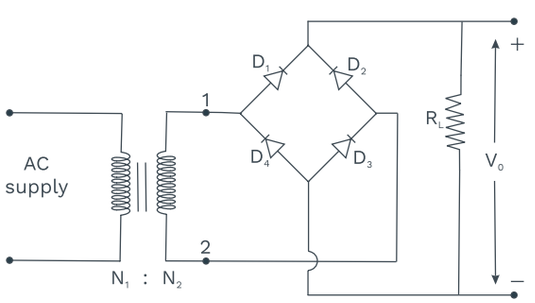



**Fig. 2.22 Circuit Diagram of Bridge Rectifier**

**Working**

- To understand the working of bridge rectifier, consider that an input AC sine wave of angular frequency (w) is applied to the primary side of the transformer.

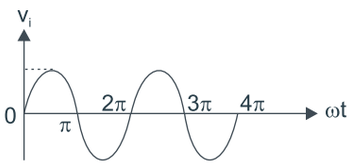



**Fig. 2.23 Input Waveform**

- Depending on whether the transformer is step up or step down, voltage is obtained across terminals 1, 2 of the transformer on secondary side.

**Positive half cycle**

- During the positive half cycle, the diodes $D_1$ and $D_3$ are in forward biased, i.e., $D_1$ and $D_3$ are in ON state, whereas $D_2$ and $D_4$ are in reverse biased, i.e., in OFF state.
- The diodes D1 and D3 are represented by a short circuit, whereas $D_2$ and $D_4$ are represented by an open circuit.

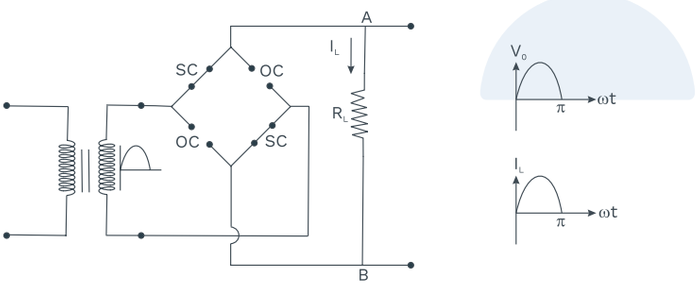



**Fig. 2.24 During Positive Half Cycle of Bridge Rectifier**

- Current flows as shown in above diagram from A to B.

**Negative half cycle**

- During the negative half cycle, the diodes $D_2$ and $D_4$ are in forward biased, whereas diode $D_1$ and $D_3$ are in reverse biased.
- $D_2$ and $D_4$ are represented by short circuit.

  $D_1$ and $D_3$ are represented by open circuit.

**Rack Your Brain**

Why are four diodes always connected in a bridge like fashion in a bridge rectifier?

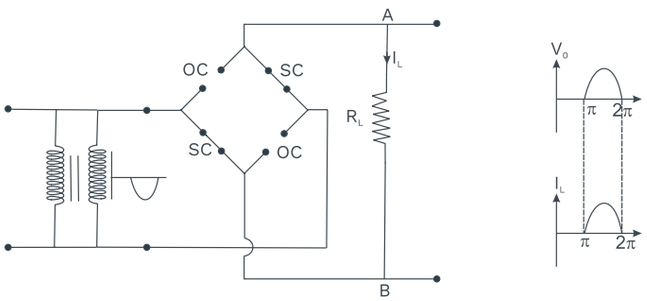



**Fig. 2.25 During Negative Half Cycle of Bridge Rectifier**

Again current flows from A to B, that is in the same direction as in previous case.

**Output waveforms**

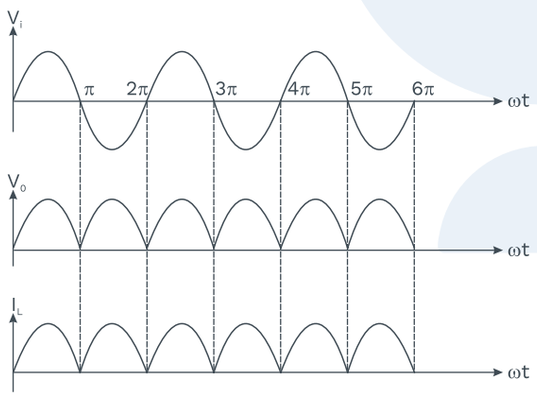



**Fig. 2.26 Output Waveforms**

**Voltage and current across the load resistance**

All the equations are same as for full wave rectifier except some minor changes.

**1) DC output current**

$$I_{DC} = \frac{2I_m}{\pi}$$

**2) RMS output current**

$$I_{rms} = \frac{I_m}{\sqrt{2}}$$

**3) AC rms output current**

$$I'_{rms} = \sqrt{I^2_{rms} - I^2_{DC}}$$

**Rack Your Brain**

Will diode conduct? If anode voltage is just equal to cathode voltage, will a diode conduct?

**4) Ripple factor**

R = 0.483

**5) Input power**

Since 4 diodes are there, and in one half cycle two diodes are in conducting state.

If forward resistance of each diode is $R_F$.

Then input power, $P_i = I^2_{rms} (2R_F + R_L)$

**6) Efficiency**

Similar to input power, efficiency is given by

$$\%\,\eta = 81 \times \frac{R_L}{2R_F + R_L}\,\%$$

**7) Voltage regulation**

$$\%\,\text{regulation} = \frac{R}{R_L} \times 100\%$$

**8) Peak inverse voltage**

$PIV = V_m$

**9) Transformer utilisation factor**

$$TUF = \frac{\text{DC output power}}{\text{AC rating of secondary winding}}$$

$$TUF = \frac{I^2_{DC} R_L}{\frac{V_m}{\sqrt{2}} \times \frac{I_m}{\sqrt{2}}}$$

$$TUF = \frac{8}{\pi^2} \frac{R_L}{(2R_F + R_L)}$$

$$TUF = \frac{0.81R_L}{(2R_F + R_L)}$$

$TUF_{max} = 0.81$

**10) Output frequency**

$f_0 = 2f_i$

**11) Angle of conduction**

Since, diodes are practical, and they always have some non-ideality, overall conduction angle is 360°– 4θ.

Where, $\theta = \sin^{-1}\left(\frac{2V_\gamma}{V_m}\right)$

**12) Peak factor**

$$PF = \frac{I_m}{I_{rms}} = 1.41$$

**13) Form factor**

$$FF = \frac{I_{rms}}{I_{DC}} = 1.11$$

**Important points regarding bridge rectifier**

- Transformer utilisation factor (TUF) is greater than the full wave rectifier.
- Peak inverse voltage is less than the full wave rectifier.

**Example 5:**

A bridge rectifier is used for rectification across load resistance of 1 kΩ. If If all the four diodes have a resistance of resistance of 50 Ω each, then calculate the efficiency.

**Sol:**

% Efficiency

$$\%\,\eta = \frac{81 \cdot R_L}{2R_F + R_L}$$

$$\%\,\eta = \frac{81 \times 1000}{100 + 1000}$$

$$\%\,\eta = \frac{81 \times 1000}{1100}$$

$$\%\,\eta = 72.27\%$$

**Rectifier circuit using filters**

- We had studied rectifier circuits using diode, where transformer and load resistor give the output as shown below.

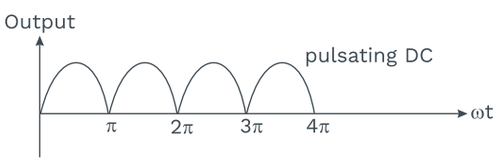



**Fig. 2.27 Waveform of Bridge Rectifier**

- We can clearly see that the output is not a pure DC; it contains both AC and DC components.

- The power supply required by an electronic circuit should be pure DC, otherwise, it creates unwanted noise in the complex electronic circuit.
- So, the next question that arises is how to reduce this AC so that output should be pure DC.
- In other words, how to reduce the ripple factor to the minimum possible value. The answer is rectifier using filter.

**Filter**

Filter is an electrical network which selects a particular range of frequency to pass and other range of frequency to reject depending upon the values of elements being used.

**Rectifier circuit using inductive filter**

In an inductive filter, just before load resistance ($R_L$) an inductor is connected as shown below.

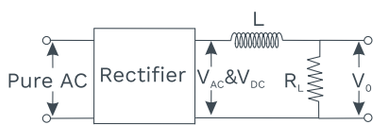



**Fig. 2.28 Rectifier with Inductive Filter**

High value of inductor should be selected as compared to $R_L$.

$w_oL >> R_L$

$X_L >> R_L$

**Working**

When $V_{AC}$ + $V_{DC}$ appears, inductor acts like a short circuit for DC, so the whole DC voltage will appear across load resistance only.

- **$V_{DC}$:** Since the inductor behaves like a short circuit for DC, so the whole DC voltage will appear across load resistance.
- **$V_{AC}$:** In case, unwanted AC signal appears across output, since inductor and resistor are behaving like a voltage divider network.

**Grey Matter Alert!**

Inductor is always used in series with the rectifier circuit.

**Grey Matter Alert!**

Without using filter circuit, we can never get pure DC signal.

$$V_{R_L} = \frac{V_{AC} \times R_L}{R_L + X_L}$$

Since $X_L >> R_L$

So $V_{R_L} << 1$

So, the output voltage will mainly contain DC voltage only.

**Ripple factor**

Ripple factor of a filter rectifier circuit is given by

$$R = \frac{R_L}{3\sqrt{2}\,\omega_0 L}$$

**Rectifier circuit using capacitive filter**

- In the capacitive filter, capacitive element is connected in parallel with the load resistance.

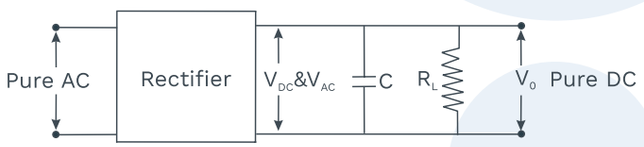



**Fig. 2.29 Rectifier with Capacitive Filter**

- Capacitor of large value should be connected, so that its reactance is smaller than $R_L$.

$X_C << R_L$

$$R_L >> \frac{1}{\omega_0 C}$$

**Working**

When $I_{DC} + I_{AC}$ appears across the output.

- **$I_{DC}$:** Since the capacitor blocks DC component, capacitor acts like an open circuit for DC.
- **$I_{AC}$:** Since the capacitor reactance is very small as compared to load resistance $R_L$, so minimum current flows through $R_L$.

$$I_{R_L} = I_{AC} \times \frac{X_C}{R_L + X_C}$$

**Rack Your Brain**

What will happen if we do not use a filter circuit in power supply network?

Since $X_C << R_L$
So, $I_{R_L}$ is very small

**Ripple factor**

Ripple factor of rectifier with capacitive filter is given by-

$$R = \frac{1}{4\sqrt{3}\, f_0\, CR_L}$$

Where, $f_0$ is the supply frequency

**Rectifier using LC filter**

- Like inductive and capacitive filter in the LC filter inductor is connected in series, and capacitor is connected in parallel.
- Ripple factor of LC filter is given by

$$R = \frac{2}{\sqrt{3}} \frac{|X_C|}{|X_L|}$$

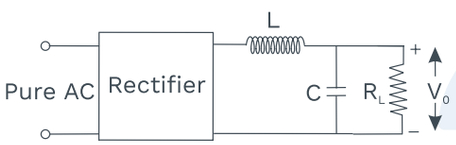



**Fig. 2.30 LC Filter**

**Diode Circuit**

In the previous section of diode, we studied in detail about rectifiers and observed how it is working, when a sinusoidal input is given.

We also studied about load line characteristics points on diode characteristics.

In this section, we will perform AC and DC analysis with small signal equivalent of diode.

We know that diode in forward biased condition has characteristics as shown below-

**Grey Matter Alert!**

Capacitor is always used in parallel with the rectifier circuit.

**Rack Your Brain**

Why all the diodes usually are practical?

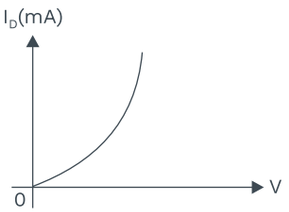



**Fig. 2.31 Practical Diode Characteristics**

The diode connected in series with DC source

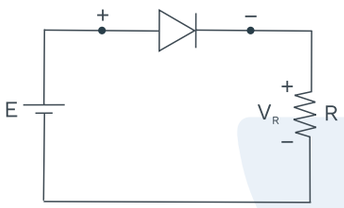



**Fig. 2.32 Series Diode Circuit**

Current through the diode is given by

$$I = \frac{E - V_R}{R}$$

Load line characteristic is given by

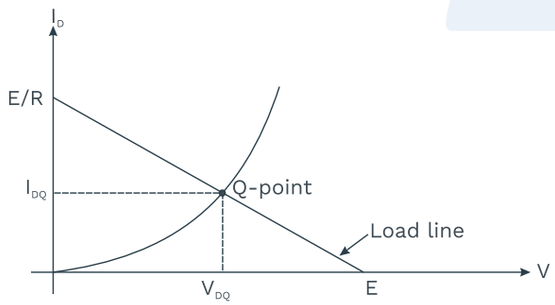



**Fig. 2.33 Load Line Characteristics**

In this section, we are going to study diode equivalent circuit with respect to DC and (AC + DC) input.

**Diode circuit with respect to some DC input**

In this section, we will study how an ideal and practical diode will behave in complicated circuit combinations.

**Grey Matter Alert!**

Generally LC filter is used in power supply circuit.

**Ideal Diode**

An ideal diode in forward biased (FB) is just replaced by a short circuit, and in reverse biased (RB), it is just replaced by an open circuit.

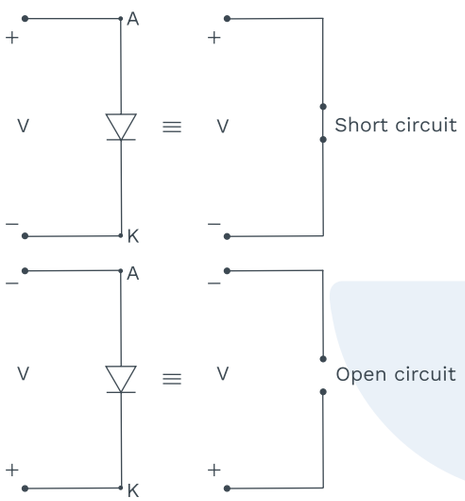



**Fig. 2.34 Representation of Ideal diode in FB and RB**

**Steps for solving diode circuits problem**

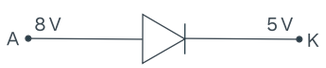
**Step 1:** Assume the diode as ideal diode, if cut-in voltage is not given.

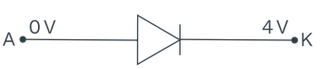
**Step 2:** Check the numerical value of the anode (A) and cathode (K) terminal voltage of the diode. If anode voltage > cathode voltage, the diode is in forward biased otherwise reverse biased.

Remember, the numerical value of voltage along with sign of polarity is important.

**Example 6:**

Check whether the following diodes are ON or OFF and represent it by an open circuit or a short circuit.

**a)** 


**b)** 

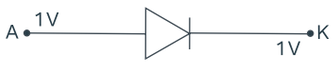

**Rack Your Brain**

Why every electronic circuit requires DC power supply to work?

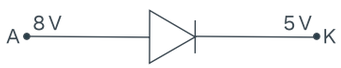
**c)** 


**d)** 


**e)** 


**f)** 


**Sol:**

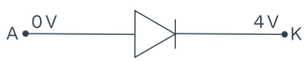
**a)** 


Since anode is connected to 8V, and cathode is connected to 5V. Diode is forward biased and it is represented by a short circuit.







**b)** 


Anode voltage is less than cathode voltage. Therefore, the diode is reverse biased, and it is represented by an open circuit.







**c)** 


Anode voltage is 0V and cathode voltage is −5 V.

Since 0V > (−5 V)

⇒Anode voltage is greater than cathode voltage, so diode is forward biased, and it is represented by a short circuit.







**d)** 


Anode voltage is −5V and cathode voltage is 0 V.

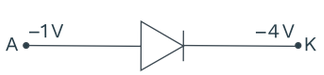
Since, −5 V < 0 V

⇒ Anode voltage is less than cathode voltage. Therefore, the diode is reverse biased, and it is represented by an open circuit.







**e)** 

Anode is connected to –1 V, and cathode is connected to –4 V

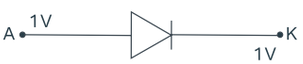
Since, –1 V > –4 V

⇒ Anode voltage is greater than cathode voltage.

⇒ Diode is forward biased. It is replaced by a short circuit.




**f)**




Cathode as well as anode is connected to 1V. Diode is reverse biased and it is represented by an open circuit




**Important note:**

A diode is said to be in forward biased only when the anode voltage is greater than cathode voltage.

**Step 3:** If it is not possible to find anode and cathode voltage of the diode, consider all the diodes to be in forward biased and find the current through the diode using simple KVL and KCL equations.

- If the current is positive, the diode is said to be in forward biased.
- If current is negative, the diode is said to be in reverse biased.

Let us solve some examples in order to understand the concept clearly.

**Example 7:**

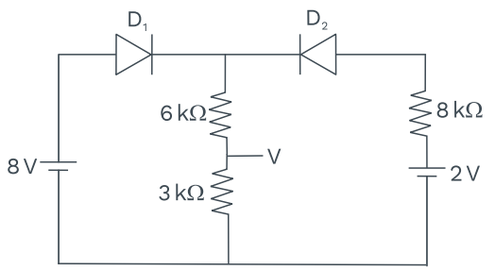
Find the value of V for the given circuit as shown in the figure below, and check whether diodes are in ON or OFF state?

**Rack Your Brain**

How diode is an useful element in communication system?

**Grey Matter Alert!**

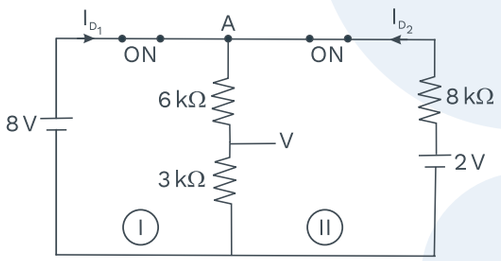
Q point is the most stable point in the characteristics of diode.




**Sol:**

As it can be clearly seen in the above circuit that cathode voltage of diode cannot be determined.

Assume diode $D_1$ and $D_2$ are in forward biased, so they are replaced by a short circuit.




Applying KCL at node A, we get-

$$I_{D_2} + I_{D_1} = \frac{8}{9} \text{ mA}$$

Applying KVL in loop II, we get-

$$8 + 8kI_{D2} - 2 = 0$$

$$I_{D_2} = \frac{-6}{8k}$$

$$I_{D_2} = \frac{-3}{4} \text{ mA}$$

Since $I_{D_2}$ is negative, which indicates that $D_2$ is in OFF condition. So, our assumption is incorrect and must be represented by an open circuit, i.e., $I_{D_2} = 0$.

Since, $I_{D_2} + I_{D_1} = \frac{8}{9}$ mA

$$I_{D_1} = \frac{8}{9}\,mA \qquad \because I_{D_2} = 0$$

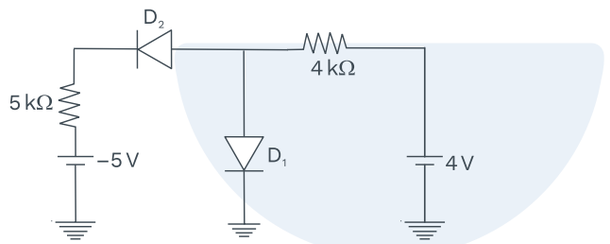
$$V = \frac{8}{9} \times 3$$

$$V = \frac{8}{3}\ V$$

Now, we can say that diode $D_1$ is in ON state and $D_2$ is in OFF state.

**Example 8:**

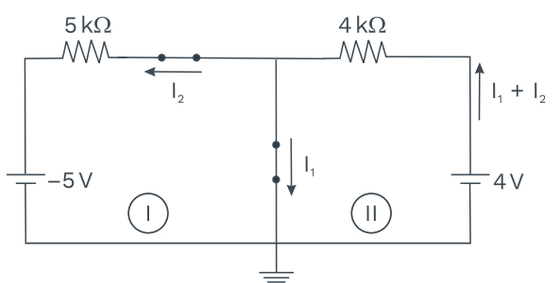
Consider the circuit shown below to find whether the diode is in ON or OFF state and also find the current through diodes.




**Sol:**

Cathode voltage of diode $D_1$ is grounded, but we can't say anything about the anode voltage, so diode states cannot be predicted.

Consider both the diodes to be in ON state.




Applying KVL in loop I,

$-5 + 5kI_2 = 0$

$I_2 = 1\ mA$

Applying KVL in loop II,

$4 - (I_1 + I_2)\ 4k = 0$

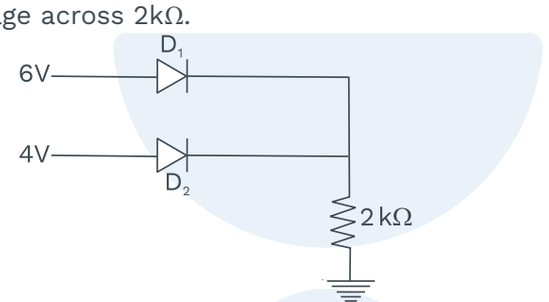
$I_1 + I_2 = 1$

$I_1 = 1–1$

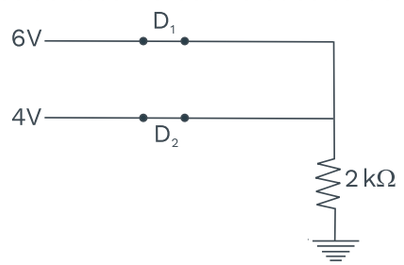
$I_1 = 0$

Since $I_1 = 0$, diode $D_1$ is in OFF state.
Now, we can say that diode $D_1$ is in OFF state and $D_2$ is in ON state, i.e., $I_1 = 0$ and $I_2 = 1$ mA.

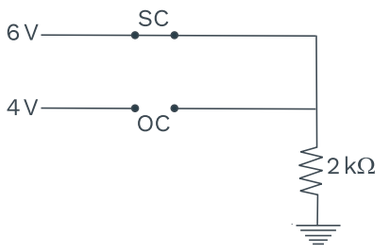
**Example 9:**
For the circuit shown blow, find which diode is ON and find current through each diode and voltage across 2kΩ.




**Sol:**
The anode voltage of each diode is positive, but we cannot say anything about the cathode voltage. Let us assume both diodes to be in ON state.




We know that voltage is equal in parallel combination, so above case is not possible.
When two diodes are connected in parallel, the diode whose anode voltage is more will be in ON state. So, $D_1$ will be ON and $D_2$ will be OFF. Replace $D_1$ by a short circuit and $D_2$ by an open circuit.




Current through 2 kΩ resistance is

$$I = \frac{6}{2k}$$

I = 3 mA

$V_{2kW} = 3 \times 2 = 6$ V

**Practical Diode**

Practical diodes generally require some minimum cut-in voltage for their operation.

Practical diodes are represented by cut-in voltage.

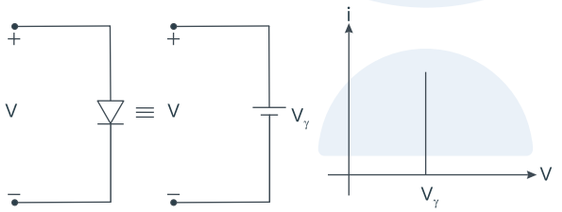
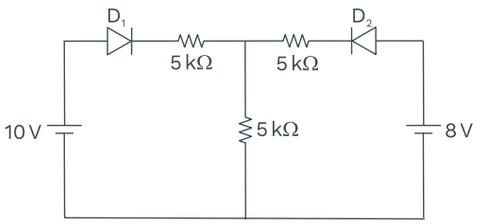



**Fig. 2.35 Representation and Characteristics of Practical Diode.**

**Example 10:**

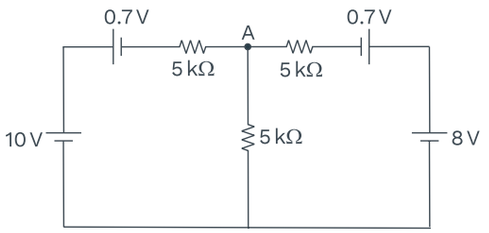
A circuit with practical diodes is as shown in figure below. Find the current through the diodes

(Assume cut-in voltage of diode to be 0.7V).



**Sol:**

Assume both the diodes to be in forward biased, and replace the diodes by cut-in voltage of 0.7 V.




Applying KCL at node A,

$$\frac{V_A - 10 + 0.7}{5k} + \frac{V_A}{5k} + \frac{V_A - 8 + 0.7}{5k} = 0$$

$$V_A - 9.3 + V_A + V_A - 7.3 = 0$$

$$3V_A = 16.6$$

$$V_A = 5.53 \text{ V}$$

Applying KCL, we get-

$$I_{D_1} = \frac{10 - 0.7 - 5.53}{5k}$$

$$I_{D_1} = 0.754 \text{ mA}$$

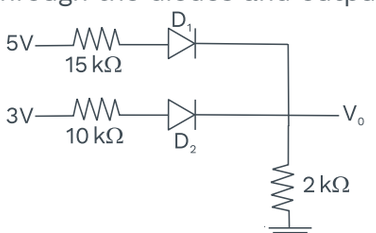
$$\text{and } I_{D_2} = \frac{8 - 0.7 - 5.53}{5k}$$

$$I_{D_2} = 0.354 \text{ mA}.$$

Current through diodes are $I_{D_1} = 0.754$ mA and $I_{D_2} = 0.354$ mA

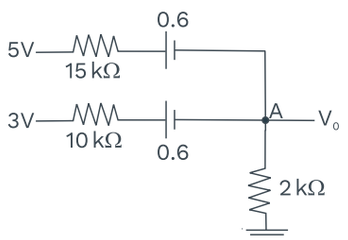
**Example 11:**

Consider the following circuit with practical diodes with cut-in voltage of 0.6V. Find the current through the diodes and output voltage.



**Sol:**

Assume both the diodes to be in forward biased. Replace each diode by cut-in voltage of 0.6 V.




Applying KCL at node A,

$$\frac{V_0 - 5 + 0.6}{15k} + \frac{V_0 - 3 + 0.6}{10k} + \frac{V_0}{2k} = 0$$

$$\frac{2(V_0 - 4.4) + 3(V_0 - 2.4) + 15V_0}{30k} = 0$$

$$2V_0 - 8.8 + 3V_0 - 7.2 + 15V_0 = 0$$

$$20V_0 = 16$$

$$V_0 = \frac{16}{20} = 0.8\,V$$

$$V_0 = 0.8\ V$$

Current through diode $D_1$ is

$$I_{D_1} = \frac{5 - 0.6 - 0.8}{15k}$$

$$I_{D_1} = \frac{3.6}{15k}$$

$$I_{D_1} = 0.24\,mA$$

Current through diode $D_2$ is

$$I_{D_2} = \frac{3 - 0.6 - 0.8}{10k}$$

$$I_{D_2} = 0.16\ mA$$

Current through the diode is given by

$$I_{D_1} = 0.24\,mA$$

$$I_{D_2} = 0.16\ mA$$

### Small Signal Model of Diode

- After studying in detail a diode circuit with DC input, we will study a diode circuit with AC inputs.
- Diode is a non-linear element, and non-linear elements are used for various applications such as small signal amplification.
  Non-linear elements use DC as the power supply, but the main signal is usually AC.

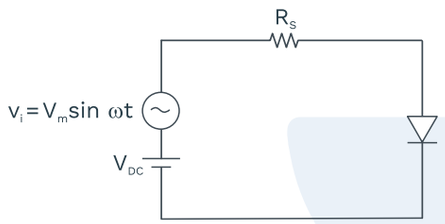



**Fig. 2.36 Diode Circuit with AC+DC Input**

- **AC Analysis**
  Replace DC voltage source by a short circuit.

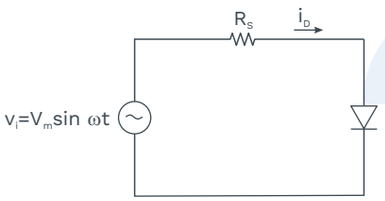



**Fig. 2.37 AC Analysis**

In this, the case current through the diode is $i_D = i_{AC}$ and voltage $V_D = V_{AC}$.

- **DC Analysis**
  Replace AC signal by a short circuit, and the circuit can be redrawn as shown below-

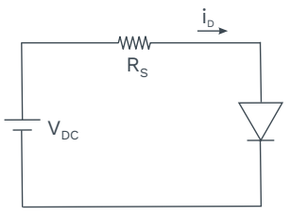
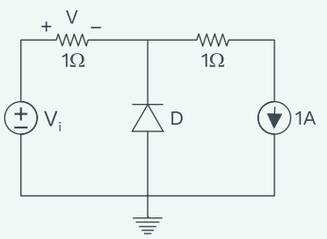



**Fig. 2.38 DC Analysis**

**Previous Years' Questions**

In the circuit below, the diode is ideal, the voltage V is given by




**a)** $\min(V_i, 1)$
**b)** $\max(V_i, 1)$
**c)** $\min(-V_i, 1)$
**d)** $\max(-V_i, 1)$

**Grey Matter Alert!**

For a very small range, non-linear diode is represented by linear resistance.

Current through the diode is $i_D = i_{DC}$

Voltage $V_D = V_{DC}$

Total current $i_D = i_{AC} + i_{DC}$

Total voltage $V_D = V_{DC} + V_{AC}$

Current through the diode, $I = I_S e^{V_D/\eta V_T}$

$$i_{AC} + i_{DC} = I_S e^{(V_{DC}+V_{AC})/\eta V_T}$$

$$= I_S e^{V_{DC}/\eta V_T} e^{V_{AC}/\eta V_T}$$

But $i_{DC} = I_S e^{V_{DC}/\eta V_T}$

$$i_{AC} + i_{DC} = i_{DC} e^{V_{AC}/\eta V_T}$$

Using Taylor series expansion,

$$e^{V_{AC}/\eta V_T} = \left[1 + \frac{V_{AC}}{\eta V_T}\right]$$

$$i_{AC} + i_{DC} = i_{DC}\left[1 + \frac{V_{AC}}{\eta V_T}\right]$$

So, $i_{AC} = i_{DC} \frac{V_{AC}}{\eta V_T}$

The above expression is the small signal AC current.

and AC resistance is given by $\frac{V_{AC}}{i_{AC}}$.

$$R_F = \frac{\eta V_T}{i_{DC}}$$

$R_F$ = Dynamic resistance of diode

So, diode is represented by a resistance $R_F$

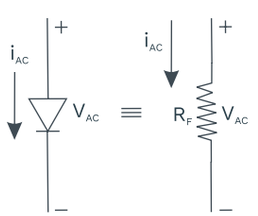



**Fig. 2.39 Representation of Diode**

$$R_F = \frac{\eta V_T}{i_{DC}}$$

It is a linear relation, so it is only valid for small signal.

It is also called a piecewise linear relation.

## CLIPPER

- As we have already seen in the previous applications of diode, the diode conducts only when it is in forward biased, which can be used to change the shape of a waveform, i.e., wave shaping circuit.

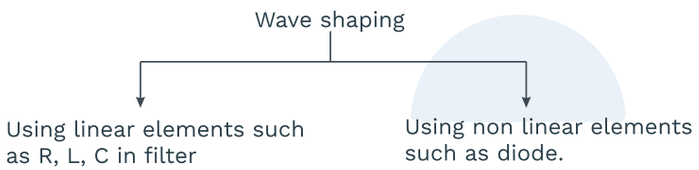
### Wave shaping of waveform

- The process of changing the shape of a waveform using proper electronic circuitry is called wave shaping.
- There are various ways by which the shape of the waveform can be changed with the use of electrical components such as diode, transistor as non linear elements and resistor, capacitor and inductor as linear elements.

**Previous Years' Questions**

For small signal ac operation, a practical forward biased diode can be modelled as

**a)** a resistance and a capacitance
**b)** an ideal diode and resistance in parallel
**c)** a resistance and an ideal diode in series
**d)** a resistance




### Clipper

It is used to cut or remove a portion of any signal without changing the remaining signal and the circuit which performs this function is known as clipper circuit.

### Why should we use clippers?

- Clipper has found number of applications in signal processing. Sometimes it is required to cut the undesired portion of the waveform such as in communication system.
  Let a message signal (m(t)) is shown in the figure.

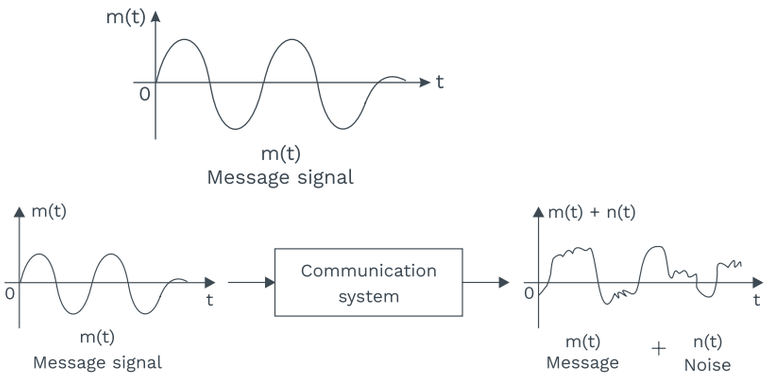



**Fig. 2.40 Signal Transmission Through Channel**

- Due to noise in the channel, the message signal gets distorted significantly. In this case, we can recover the message signal by using a clipper circuit.
- Sometimes the magnitude of input signal is large that may damage the device which cannot withstand such high magnitude of voltage or current, so clipper can be used to limit the amplitude of input signal.

**Grey Matter Alert!**

Clipper may be called series or shunt depending upon the way it is connected with the input voltage.

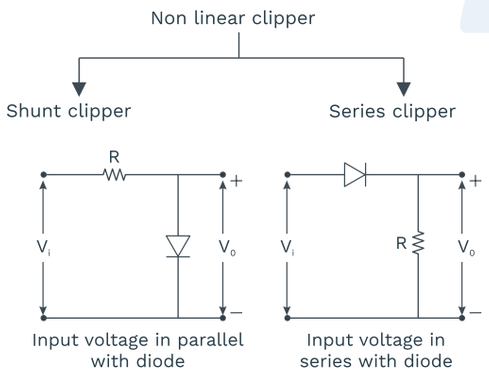



**Fig. 2.41 Clipper Circuits**

**Grey Matter Alert!**

Clipper circuit is very useful for removing the noise in modulated signal in the communication system.

**Shunt clipper**

The components needed in shunt clippers are

- Resistance for current limiting
- Diode
- Reference voltage source: the voltage at which we want to clip the waveform.

Suppose we have a sine wave of peak amplitude 10 V.




**Fig. 2.42 Input Signal**

We want to clip the waveform above 5V. So, reference voltage, $V_R$ = 5V.

### Circuit diagram

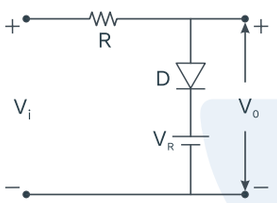



**Fig. 2.43 Circuit Diagram of Shunt Clipper**

### Working

- When input voltage $V_i > V_R$, diode is in forward bias, and when $V_i \leq V_R$, diode is in reverse bias.
- In the forward bias entire voltage waveform is passed upto reference voltage. In the reverse bias entire voltage waveform is blocked.
  When $V_i > V_R$, replace diode by a short circuit

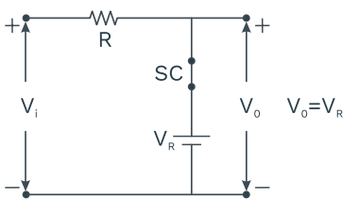



**Fig. 2.44 Shunt Clipper, When $V_i > V_R$**

When $V_i < V_R$, replace diode by an open circuit

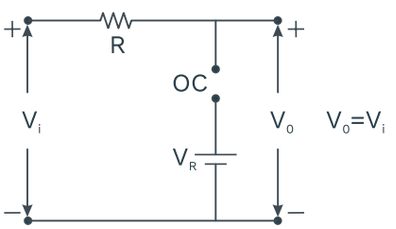
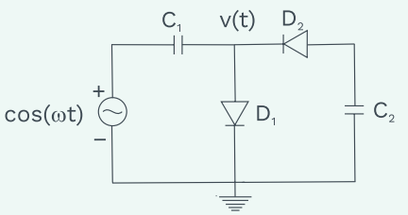



**Fig. 2.45 Shunt Clipper, When $V_i < V_R$**

### Previous Years' Questions

The diodes and capacitors in the circuit shown are ideal. The voltage v(t) across the diode $D_1$ is




**a)** cos(ωt) – 1
**b)** sin(ωt)
**c)** 1 – cos(ωt)
**d)** 1 – sin(ωt)

**Waveforms**

Consider a sine wave of amplitude 10 V and $V_R$ = 5 V.

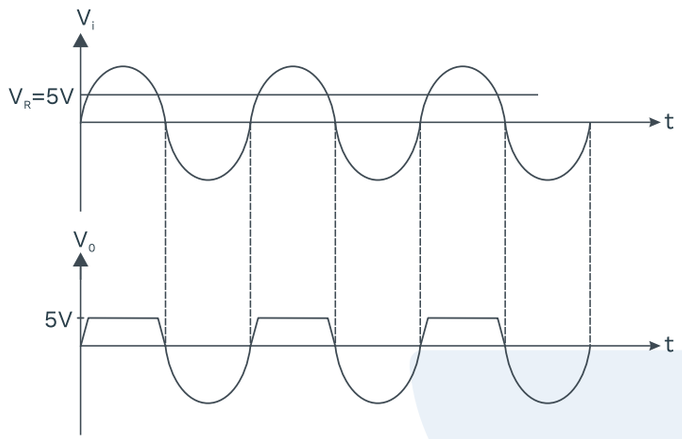



**Fig. 2.46 Output Waveform of Clipper Circuit**

**Transfer characteristics**

It is drawn between input and output.

$V_i > V_R \Rightarrow V_0 = V_R$

$V_i < V_R \Rightarrow V_0 = V_i$

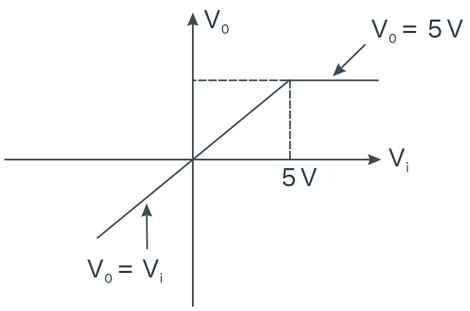



**Fig. 2.47 Transfer Characteristics**

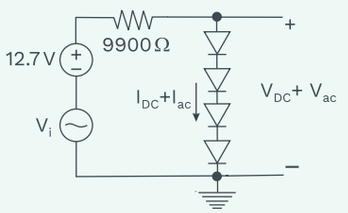
**Example 12:**

A circuit is shown below, draw output waveform and transfer characteristics. (Assume $V_R$= 3 V and diode to be practical)

**Previous Years' Questions**

**Statement for Linked Answer Questions**

In the circuit shown, assume that the voltage drop across a forward biased diode is 0.7 V. The thermal voltage $V_t$ = kT/q = 25 mV. The small signal input v(t) = $V_p$ cos(ωt) where $V_p$ =100 mV.

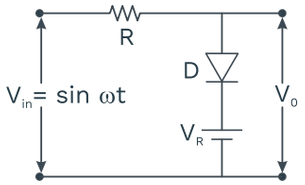



**1)** The bias current $I_{DC}$ through the diodes is

**a)** 1 mA
**b)** 1.28 mA
**c)** 1.5 mA
**d)** 2 mA

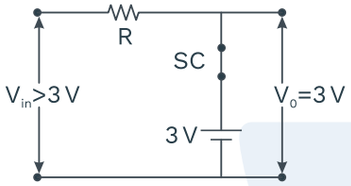
**2)** The ac output voltage $V_{ac}$ is

**a)** 0.25 cos(ωt) mV
**b)** 1 cos(ωt) mV
**c)** 2 cos(ωt) mV
**d)** 22 cos(ωt) mV

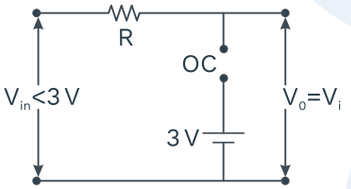



**Sol:**

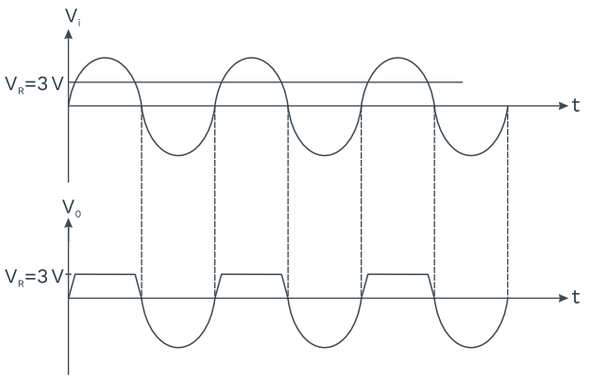
When $V_{in}$> 3 V, diode is in forward biased and replaced by a short circuit.




When $V_{in}$< 3 V, diode is in reverse biased and replaced by an open circuit.

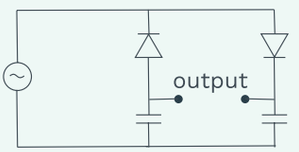



**Waveform:**



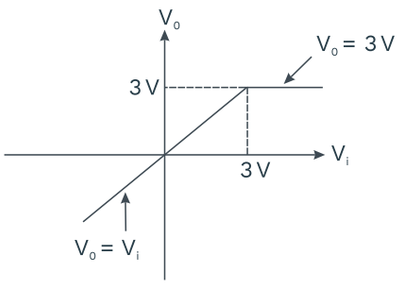

**Previous Years' Questions**

The circuit shown in the figure is best described as a



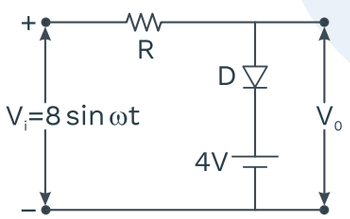

**a)** bridge rectifier
**b)** ring modulator
**c)** frequency doubler
**d)** voltage doubler

**Transfer characteristics**



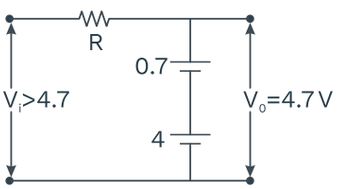

**Example 13:**

A practical diode circuit is shown below with a cut-in voltage of 0.7 V, draw output waveforms and transfer characteristics.

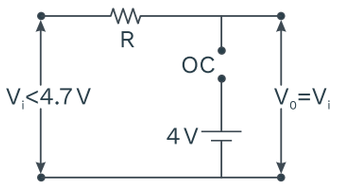



**Sol:**

When $V_i > 4.7$ V, replace diode by its cut-in voltage of 0.7 V.

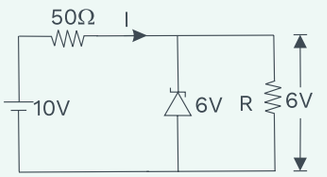



When $V_i < 4.7$ V, replaced diode by open circuit.



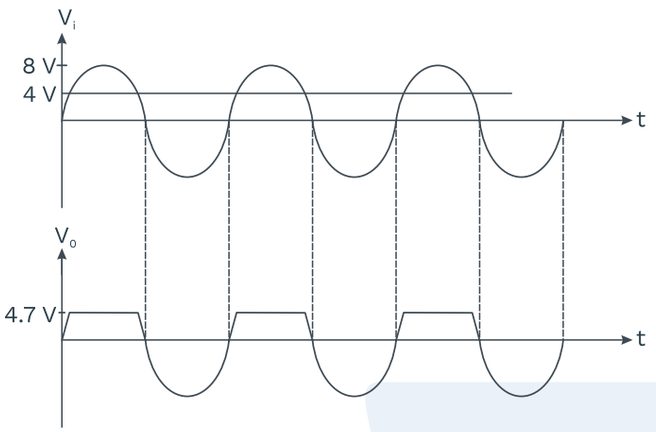

**Previous Years' Questions**

The 6V Zener diode shown below has zero Zener resistance and a knee current of 5 mA. The minimum value of R, so that the voltage across it does not fall below 6 V is-

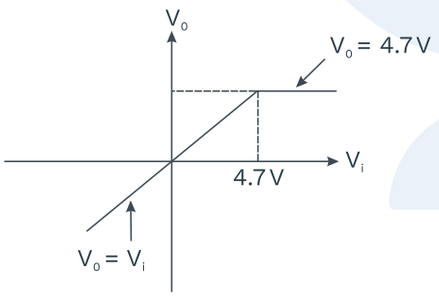



**a)** 1.2 kΩ **b)** 50 Ω
**c)** 80 Ω **d)** 0 Ω

**Waveform**




**Transfer characteristics**




**Grey Matter Alert!**

Clipper circuit always cuts the waveform but cannot reconstruct the original waveform.

**Series Clipper**

In the series clipper circuit, input voltage is in series with the diode.

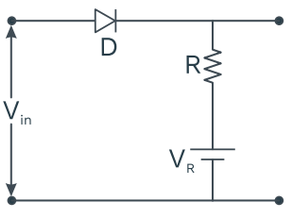



**Fig. 2.48 Series Clipper Circuit**

**Working**

When input voltage is greater than $V_R$ , i.e., $V_{in} > V_R$, diode is in forward biased and replaced by short circuit.

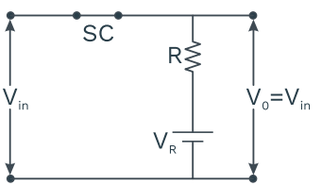



**Fig. 2.49 Series Clipper, When $V_{in} > V_R$**

When $V_{in} < V_R$, the diode is in reverse diode and replaced by open circuit.

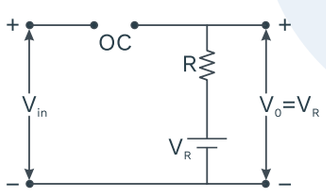



**Fig. 2.50 Series Clipper, When $V_{in} < V_R$**

**Waveform**

Consider the applied input is sine wave.

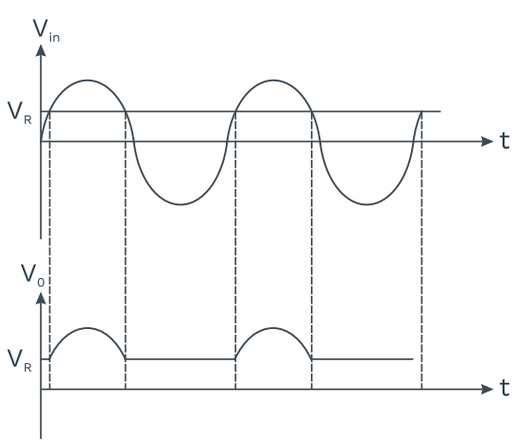



**Fig. 2.51 Input and Output Waveforms**

**Grey Matter Alert!**

Operation of series clipper is just opposite to that of shunt clipper.

**Transfer characteristics**

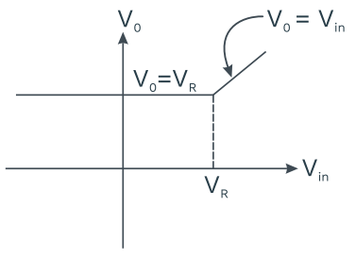
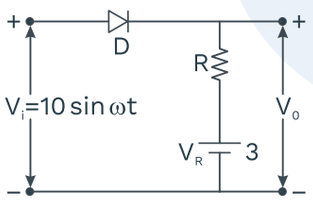



**Fig. 2.52 Transfer Characteristics**

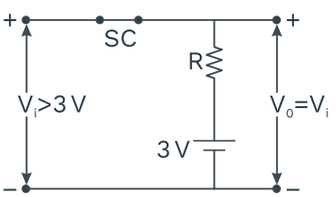
**Example 14:**

A ideal diode circuit is connected as shown in the figure below, draw output waveform and transfer characteristics.

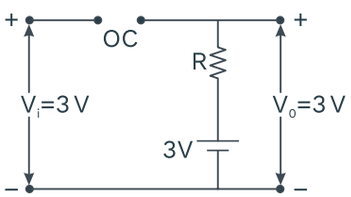



**Sol:**

When $V_i$ > 3 V, diode will be in forward bias and replaced by short circuit.



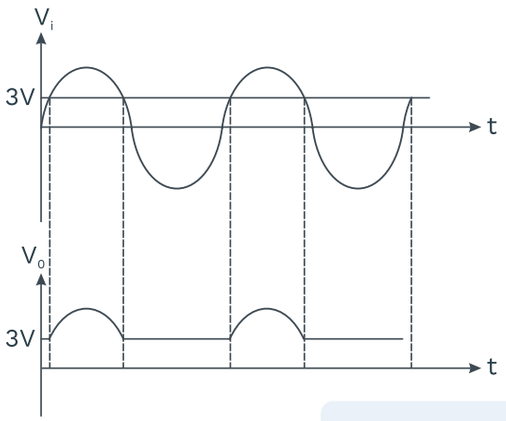

When $V_i$< 3 V, diode will be in reverse bias and replaced by open circuit.



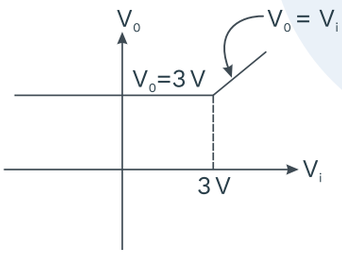

**Rack Your Brain**

Can we recover the original waveform after clipping it?

**Waveform**




**Transfer characteristics**




**Two Diode Clipper**

In a single diode clipper circuit, there is only one reference voltage $V_R$. But if we want to clip the waveform between two different reference voltages $V_{R1}$ and $V_{R2}$, then single diode clipper circuit is not enough. So, we have to use two diodes circuit in order to clip the waveform.

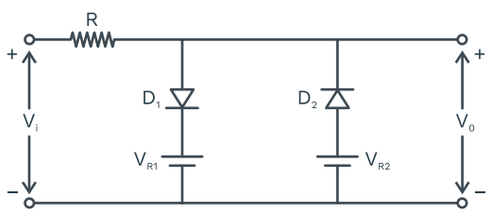


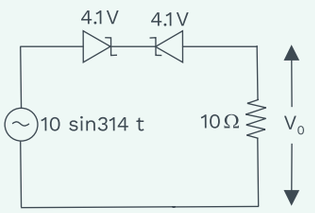

**Fig. 2.53 Two Diode Clipper Circuit**

Assumption: $V_{R1} > V_{R2}$

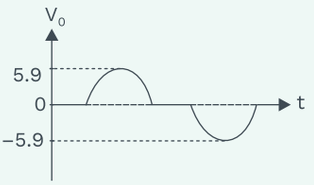
**Case 1:** When $V_i > V_{R1}$ and $V_{R2}$ diode, $D_1$ is in forward biased and diode $D_2$ is in reverse biased. $D_1$ is replaced by a short circuit and $D_2$ by an open circuit.

**Previous Years' Questions**

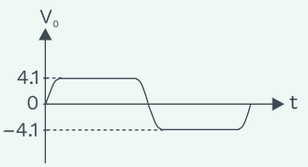
The wave shape of $V_0$ in figure is-



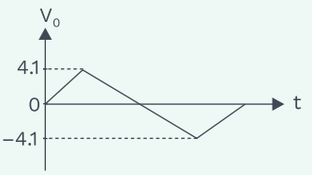

**a)**



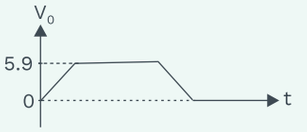

**b)**




**c)**




**d)**



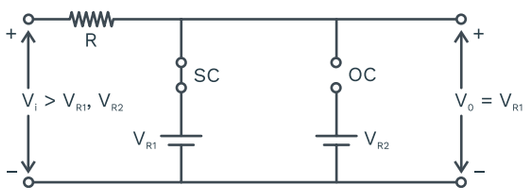



**Fig. 2.54 Clipper Circuit, When $V_i > V_{R1}, V_{R2}$**

**Case 2:** When $V_{R2} < V_i < V_{R1}$, diode $D_1$ is in reverse biased and diode $D_2$ is in reverse biased.

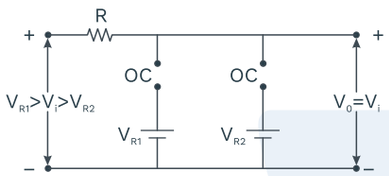



**Fig. 2.55 Clipper Circuit, When $V_{R2} < V_i < V_{R1}$**

**Case 3:** When $V_i < V_{R1}, V_{R2}$, diode $D_1$ is in reverse biased and diode $D_2$ is in forward biased.

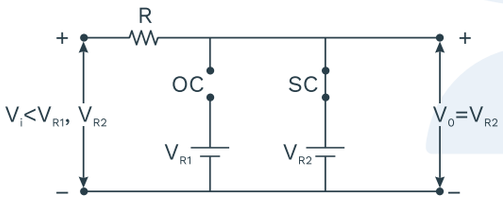


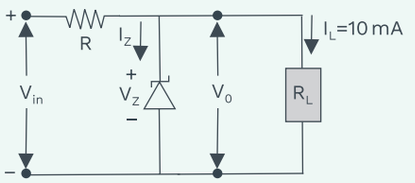

**Fig. 2.56 Clipper Circuit, When $V_i < V_{R1}, V_{R2}$**

### Clamper

- After going through the number of application of diode such as rectifiers, clippers among others, the next application of diode is to shift the waveforms up or down.
- The shifting of waveform up or down is known as clamping and the circuit which performs this operation is known as clamper circuit. But the question is why do we need clamping? In other words, why should we shift the waveform up or down?
- Clamper has a number of applications in electronic circuits.

### Previous Years' Questions

A Zener diode regulator in the figure is to be designed to meet the specification $I_Z$ = 10 mA, $V_0$ = 10 V, and $V_{in}$ varies from 30 V to 50 V. The Zener diode has $V_Z$ = 10V and $I_Z$ (knee current) = 1 mA. For satisfactory operation.




**a)** R ≤ 1800 Ω
**b)** 2000 Ω ≤ R ≤ 2200 Ω
**c)** 3700 Ω ≤ R ≤ 4000 Ω
**d)** R > 4000 Ω

Eg: Communication system.
In analog modulation scheme, the equation of amplitude modulated wave is given by
$s(t) = A_c (1 + m(t)) \cos wt$

- Here message signal is shifted up by positive clamper making the process of modulation and transmission easy.

Consider message signal m(t) as shown below.

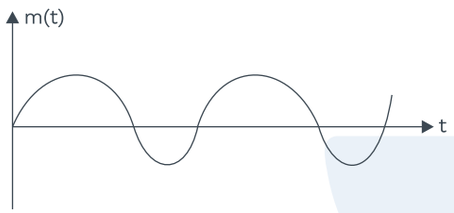



**Fig. 2.57 Message Signal**

- In the modulation, we shift the waveform and perform amplitude modulation.

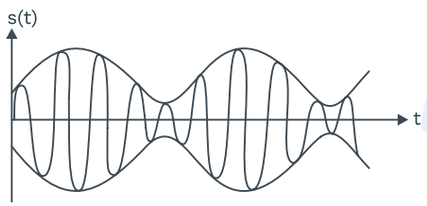
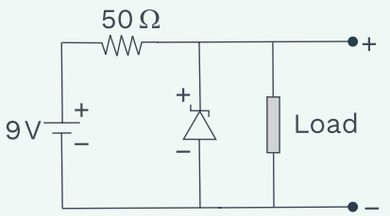



**Fig. 2.58 Modulated Signal**

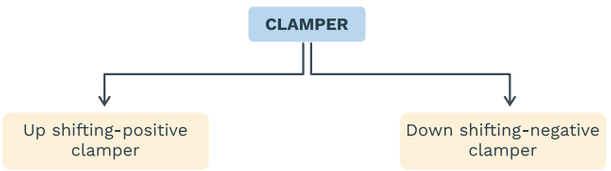
- If we do not perform the operation of adding DC, then cross modulation occurs.
- Not only in communication, but there are various applications of clamper circuit in analog signal processing circuits.
- In amplifier circuits, DC component added to signals during amplification can be removed using clamper.
- Clamper is classified based on shifting the signal up or down, Hence, it is classified as a positive and negative clamper.

**Previous Years' Questions**

A Zener diode circuit shown in below figure has a knee current of 5 mA, and a maximum allowed power dissipation of 300 mW. What are the minimum and maximum load currents that can be drawn safely from the circuit keeping the output voltage $V_0$ constant at 6 V?




**a)** 0 mA, 180 mA
**b)** 5 mA, 110 mA
**c)** 10 mA, 55 mA
**d)** 60 mA, 180 mA




**Positive Clamper**

- A positive clamper circuit is shown below:

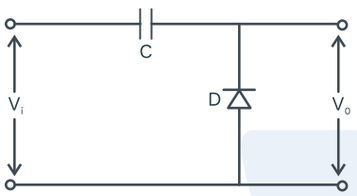



**Fig. 2.59 Positive Clamper**

- The capacitor used in the circuit stores the AC voltage in DC form, which shifts the waveform.
- Diode can be forward or reverse biased depending upon the polarity of input signal.
- Let the shifted waveform is applied at the input port and shifted waveform is obtained at the output port.
- When input is negative, diode becomes forward biased and capacitor starts charging through diode to the peak amplitude of input signal $V_m$.

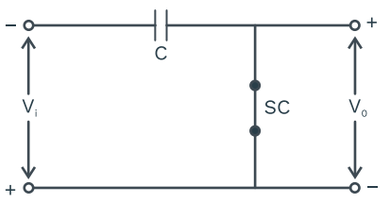
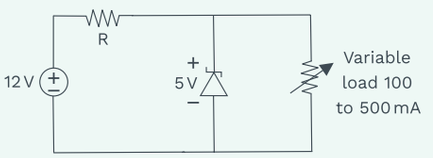



**Fig. 2.60 Working of Positive Clamper**

- When input is positive, diode becomes reverse biased and should be replaced by open circuit. Capacitor tries to discharge but as there is no path available for discharge, It remains as it is with some peak amplitude of input DC signal.

**Previous Years' Questions**

In voltage regulator shown in the figure, the load current can vary from 100 mA to 500 mA. Assuming that the Zener diode is ideal (i.e. the Zener knee current is negligibly small, and Zener resistance is zero in the breakdown region), the value of R is




**a)** 7 Ω
**b)** 70 Ω
**c)** $\frac{70}{3}\Omega$
**d)** 14Ω

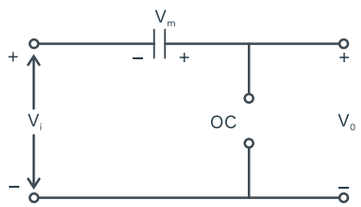



**Fig. 2.61 Working of Positive Clamper**

Applying KVL in the circuit, we get

$V_0 = V_i + V_m$

### Waveforms

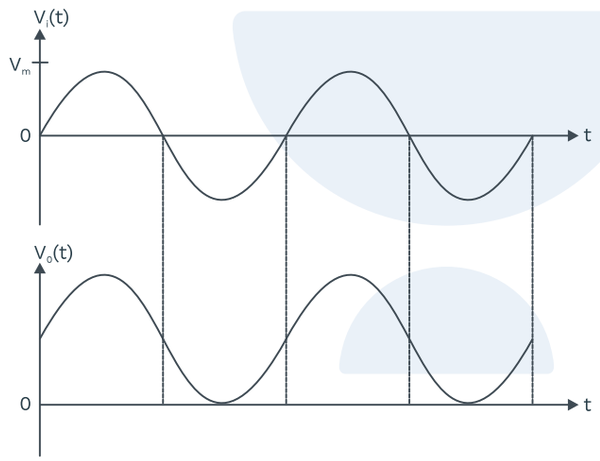



**Fig. 2.62 Waveforms of Positive Clamper**

If diode has cut-in voltage of $V_g$ (practical diode)

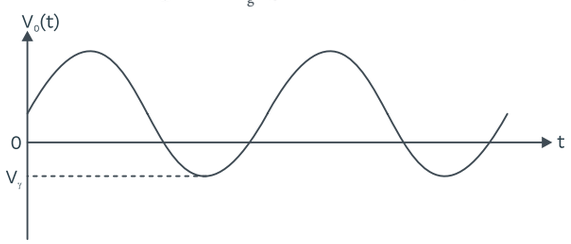



**Fig. 2.63 Output Waveform When Practical Diode is Used**

### Negative Clamper

- A circuit for negative clamper is shown below.

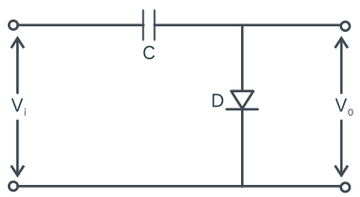



**Fig. 2.64 Negative Clamper**

- When the input signal is positive, diode becomes forward biased and should be replaced by short circuit, capacitor starts charging and charged upto maximum value of input signal.

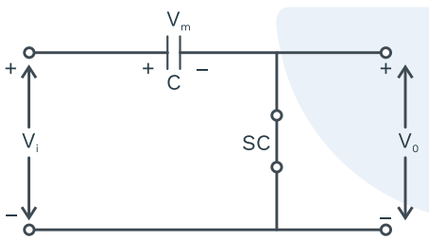



**Fig. 2.65 Negative Clamper, When Input Signal is Positive**

When the input is negative, diode becomes reverse biased and should be replaced by an open circuit, capacitor remains charged upto $V_m$.

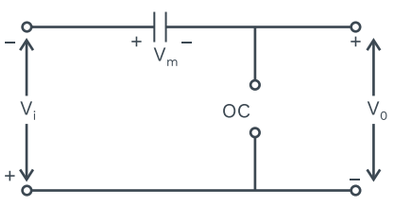



**Fig. 2.66 Negative Clamper, When Input Signal is Negative**

- Capacitor tries to discharge, but as there is no path available for discharge, it remains with voltage $V_m$.

  Applying KVL in the above circuit, we get

  $V_i - V_m - V_0 = 0$

  $V_0 = V_i - V_m$

**Grey Matter Alert!**

For getting a particular portion of waveform between two reference levels, a two-diode clipper circuit is necessary.

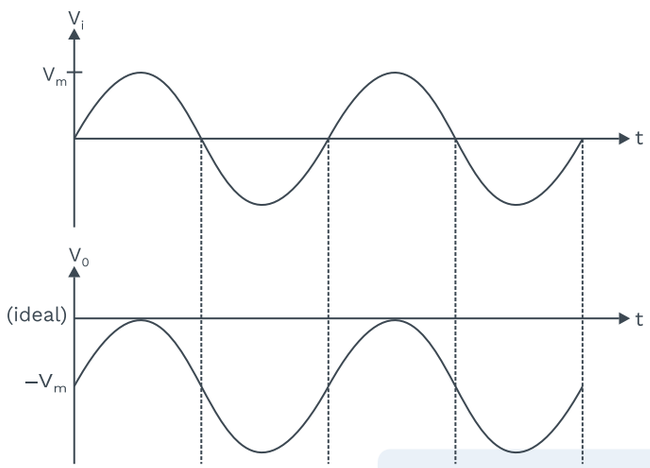



**Fig. 2.67 Waveform of Negative Clamper**

### Voltage Multiplier

- Voltage multiplier is nothing but the extended version of clamper circuit.
- In the clamper circuit, output is taken across the diode, but in the voltage multiplier circuit, output is taken across the capacitor in DC form.

  **Note:** In multiplier circuit input is AC but output is DC because it is taken across capacitor.

### Voltage doubler

- The circuit for voltage doubler is shown below.

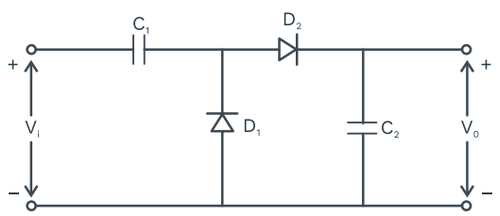



**Fig. 2.68 Voltage Doubler**

### Case 1:

- When input is negative, diode $D_1$ gets forward biased and $D_2$ gets reverse biased.
  So, $D_1$ is replaced by a short circuit, and $D_2$ is replaced by an open circuit.

**Rack Your Brain**

What is the use of clamper circuit in amplitude modulation?

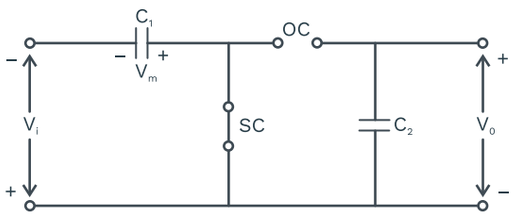



**Fig. 2.69 Working of Voltage Doubler**

Capacitor starts charging and charge upto maximum voltage $V_m$.

**Case 2:**

- When input is positive, diode $D_1$ becomes reverse biased and diode $D_2$ becomes forward biased, $D_1$ is replaced by an open circuit and $D_2$ by a short circuit.

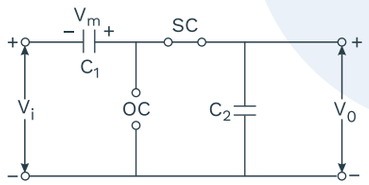



**Fig. 2.70 Working of Voltage Doubler**

Applying KVL voltage across capacitor becomes

$$V_0 = V_m + V_m$$

$$V_0 = 2V_m$$

So, output voltage is two times the peak amplitude of input. Hence, named as voltage doubler.

DC output voltage $V_0 = 2V_m$

**Voltage Tripler/Quadrupler**

Some circuit can act as a voltage doubler or voltage tripler or voltage quadrupler.

Circuit is shown below:

**Rack Your Brain**

What is the use of voltage multiplier circuit?

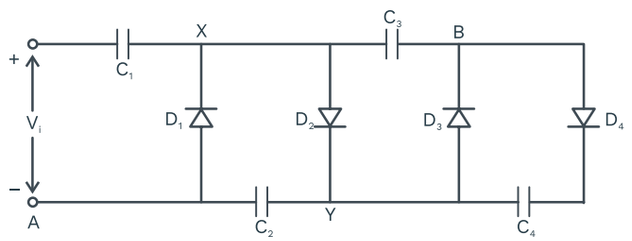



**Fig. 2.71 Voltage Tripler/Quadrupler**

**Case 1:**

When input voltage is negative diode $D_1$ becomes forward biased and capacitor $C_1$ charges upto $V_m$.

**Case 2:**

When input voltage is positive after applying KVL through loop, we get-

$$V_{XA} = V_m + V_m = 2V_m$$

In positive half cycle, voltage through $D_1$ is $2V_m$, capacitor $C_2$ starts charging through $D_2$ upto voltage $2V_m$.

**Case 3:**

Again when input voltage is negative,

$$V_{XA} = V_m - V_m = 0$$

$$V_{YA} = 2V_m$$

$D_3$ becomes forward biased, then $C_3$ charges through $D_3$upto a voltage of

$$V_{YA} - V_{XA} = 2V_m - 0 = 2V_m$$

**Case 4:**

When input voltage is positive,

$$V_{BA} = 2V_m + V_m + V_m$$

$$V_{BA} = 4V_m$$

So, output is $4V_m$ if measured across $C_4$ and $C_2$ and output is $3V_m$ if measured across $C_3$ and $C_1$.

**Rack Your Brain**

What is the meaning of xx in 78xx IC?

**Regulator Circuit**

- One of the most famous applications of diode is the regulator circuit.
- In electronic circuitry for proper operation such as amplification or switching, it requires a fixed DC voltage supply, which means that whatever may be the fluctuations in the other circuit voltages or currents, the supply voltage should be fixed.
- For this application, the diodes can be used in reverse biased condition.

**Rack Your Brain**

Why only Zener diode is used in regulator circuit and not avalanche diode?

**V–I characteristics of diode**

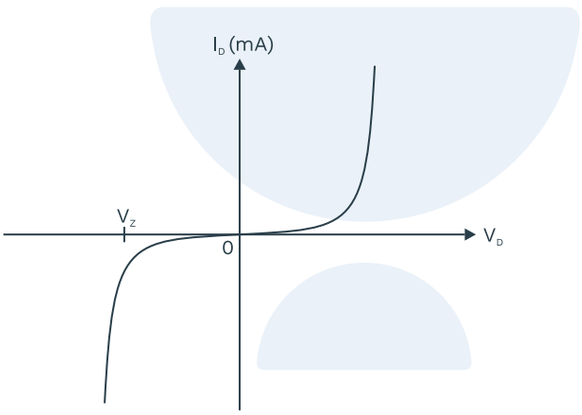



**Fig. 2.72 V–I Characteristics of Diode**

- When reverse biased voltage is applied to diode with high doping concentrations, for certain fixed value of voltage diode breakdown starts and eventually a very large amount of current starts flowing through the diode, but the value of voltage is fixed.
- This characteristics is unique in the highly doped diode popularly known as Zener diode.
- In this section our focus is on the Zener diode and its operation.
- A heavily doped reverse biased diode acts as voltage regulator.
- It should be clearly noted that voltage of regulator circuit is proportional to the doping concentration, the more the doping concentration, the less the breakdown voltage.
- Breakdown voltage is denoted by $V_Z$ and symbolic representation of zener diode is given below-

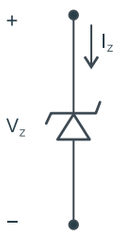



**Fig. 2.73 Symbol of Zener Diode**

### Characteristics of Zener Diode

The voltage across Zener diode is always constant irrespective of change in the load current.




**Fig. 2.74 Characteristics of Zener Diode**

The percentage regulation of ideal Zener diode is always 100%.

### Specification sheet of Zener diode

A typical specification sheet of Zener diode at room temperature is given below:

| Characteristics | Value |
|---|---|
| Zener voltage ($V_Z$) (in V) | 10 |
| Test current ($I_{ZT}$) (in mA) | 12.5 |
| Minimum dynamic resistance (Z) (in Ω) | 8.5 |
| Maximum dynamic resistance (Z) (in kΩ) | 700 |
| Maximum reverse current ($I_R$) (in A) | 10 |
| Test voltage ($V_R$) (in V) | 7.2 |
| Maximum regulator current ($I_{Z(max)}$) (in mA) | 32 |
| Typical temperature coefficient (%/°C) | 0.072 |

**Table 2.3**

**Note:**
The voltage of Zener diode is highly dependent on the temperature.

**Design of Voltage Regulator**

- In order to design a regulator circuit, the specification of Zener diode is used such as:
  $I_{Z\,(max)}$, $I_{Z\,(min)}$, $P_{Z\,(max)}$, $I_{L\,(min)}$, $I_{L\,(max)}$, $R_{L\,(min)}$ and $R_{L\,(max)}$
- Zener diode is connected to a load resistance, which is to be maintained at fixed voltage.

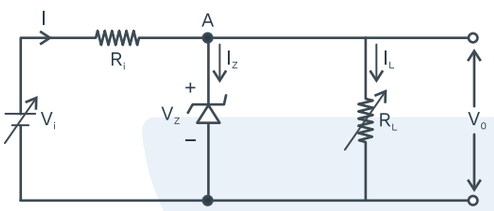



**Fig. 2.75 Voltage Regulator Circuit**

- $V_i$ = input voltage for operating.
- Input voltage supply is highly fluctuating sometimes.
  $R_i$: Input resistance

  Applying KVL in the input loop, we get-

  $$I = \frac{V_i - V_Z}{R_i}$$

  Applying KCL at node A, we get-

  $$I = I_L + I_Z$$

**Case 1:**
If input voltage is constant, I is constant.

$$I = I_{Z\,min} + I_{Lmax}$$

$$I = I_{Z\,max} + I_{L\,min}$$

When Zener current is minimum, load current is maximum, and when Zener current is maximum load current is minimum, if input voltage is constant.

**Case 2:**
If input voltage is varying, the current I is also varying

$$I_{max} = \frac{V_{imax} - V_Z}{R_i}$$

$$I_{min} = \frac{V_{imin} - V_Z}{R_i}$$

Again applying KCL at node A, we get

$$I = I_Z + I_L$$

Since, load resistance is constant and Zener breakdown voltage is also constant. So, $I_L$ is also constant. That is,

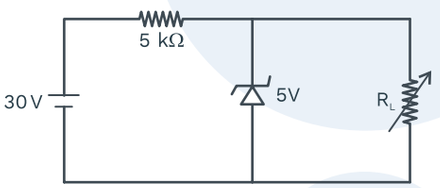
$$I_L = \frac{V_Z}{R_L} = \text{constant}$$

$$I_{max} = I_{Z\,max} + I_L$$

$$I_{min} = I_{Z\,min} + I_L$$

**Example 15:**

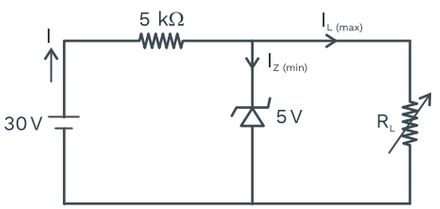
A Zener diode regulator circuit is shown below, calculate the value of minimum and maximum Zener current.




If $R_L$ is varying between 2 kW and 4 kW.

**Sol:**

Since input voltage is constant at 30 V, so input current is also constant.




Since $R_L$ = 2000 Ω (minimum), So current $I_L$ is maximum.

$$I_{Lmax} = \frac{5}{2000}$$

$$I_{L\,max} = 2.5 \text{ mA}$$

When $R_L$ = 4000 Ω (maximum)

$$I_{L\,min} = \frac{5}{4000}$$

$$I_{L\,min} = 1.25\ mA$$

Applying KVL in the input loop, we get-

$$I = \frac{30 - 5}{5k}$$

$$I = 5\ mA$$

$$I_{Z\,max} = I - I_{L\,min}$$

$$I_{Z\,max} = (5 - 1.25)\ mA$$

$$I_{Z\,max} = 3.75\ mA$$

$$I_{Z\,min} = I - I_{L\,max}$$

$$I_{Z\,min} = (5 - 2.5)\ mA$$

$$I_{Z\,min} = 2.5\ mA$$

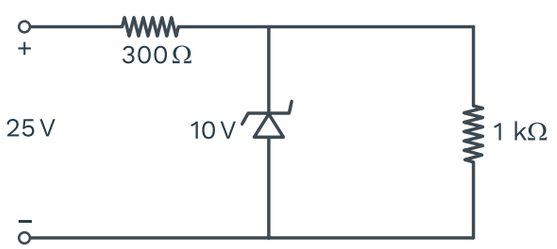
**Example 16:**
Consider the following statement with respect to Zener diode and choose the correct statement

**a)** Diode gets damaged **b)** Current is constant

**c)** Voltage is constant **d)** Current decreases

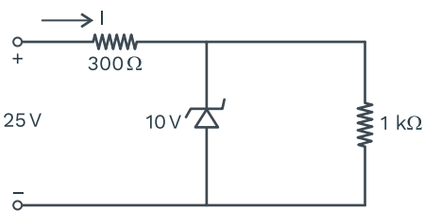
**Sol:**
After breakdown, voltage across the diode is constant.
Hence, the option c) is correct.

**Example 17:**
A Zener diode regulator circuit is shown in the figure below, the current through the Zener diode is-



**Sol:**

Applying KVL in the input loop,




$$I = \frac{25 - 10}{300}$$

$$I = \frac{15}{300}$$

$$I = 50 \text{ mA}$$

$$I_L = \frac{10}{1000} = 10 \text{ mA}$$

Current through the diode is,

$$I_z = 50 - 10$$

$$I_z = 40 \text{ mA}$$

## Chapter Summary



- Diode is a two-terminal device; terminals are anode and cathode, and diode is made up of semiconductor material.
- Current flows from anode to cathode terminal in the diode.
- Diode is the fundamental building block of any electronic circuit.
- Rectifier is the most widely used application of diodes.
- Rectifiers can be classified into 2 categories.
  - **1)** Half wave rectifier
  - **2)** Full wave rectifier
    - **i)** Centre tapped full wave rectifier
    - **ii)** Bridge type full wave rectifier.
- Ripple factor of half wave rectifier is 1.21, which means that AC component is 1.21 times the average component.
- Maximum efficiency of half wave rectifier is 40.5%.
- Angle of conduction is nothing but the time for which the diode is in ON state.
- Full wave rectifier uses two diodes which conducts in alternate half cycles.
- Ripple factor of full wave rectifier is 0.483.
- Efficiency of full wave rectifier is 81%.
- Bridge rectifier uses 4 diodes with 2 diodes conducting in every half cycle.
- Efficiency of bridge rectifier is 81%.
- Ideal diode in forward biased is replaced by a short circuit, and in reverse biased it is replaced by an open circuit.
- Practical diode in forward biased is replaced by its cut-in voltage.
- Clipper circuit is used for wave shaping of signal waveform.
- Clipper circuit cuts or removes a portion of waveform using a simple diode circuitry.
- Clipper circuit can be classified as
  - **1)** Series clipper
  - **2)** Shunt clipper

- The shift of waveform up or down is known as clamping.
- Clamper circuit is used in analog communication system to shift the message signal.
- Clamper is classified as
  **1)** Positive clamper
  **2)** Negative clamper
- Input of voltage multiplier is AC signal, but the output is DC signal.
- Regulator is a widely used application of diode.
- Zener diode is used as a voltage regulator in an electronic circuitry for providing constant DC voltage.
- Zener diode is always used in reverse biased mode.

## PRACTICE QUESTIONS

**Q1** **A rectifier is a circuit which converts**
**a) AC to DC**
**b) DC to AC**
**c) AC to AC**
**d) Rectifier is nothing but a transformer**

**Q2** **Which of the following type of rectifier has more complex circuitry?**
**a) Half wave rectifier**
**b) Full wave rectifier**
**c) Bridge rectifier**
**d) Silicon controlled rectifier**

**Q3** **A rectifier circuit uses**
**a) Only step up transformer**
**b) Only step down transformer**
**c) depending on application and value of input voltage**
**d) Centre tapped transformer**

**Q4** **A input sine voltage of 220 V RMS value is applied to a half wave rectifier through a step down transformer of turn ratios N : 1, if average value of load current is 500 mA. Find N, if load resistance is $R_L$ = 5 kΩ.**
**a) 80** **b) 75** **c) 64** **d) 50**

**Q5** **Find conduction angle of half wave rectifier, if sine wave of 50 mV peak voltage is applied and diode resistance is 50Ω with cut-in voltage 0.7 V.**
**a) 170** **b) 160** **c) 180** **d) 150**

**Q6** In a bridge rectifier circuit, 4 diodes with forward resistance 25 Ω are used. If source resistance of transformer is 100 Ω on secondary, find average value of load current if average value of DC output voltage is 150V, load resistance is 1 kΩ.

a) 130.4 mA b) 168 mA c) 150.6 mA d) 180 mA

**Q7** The output of half wave rectifier is taken across load resistance of 5 kΩ, practical resistance of diode is 20 Ω. Find practical efficiency of the rectifier.

a) 40.5 b) 40.33 c) 40.9 d) 41

**Q8** In bridge rectifier, source resistance is 100 Ω and practical resistance of diode is 30 Ω. Find the regulation, if load resistance is 4 kΩ.

a) 2 b) 3 c) 4 d) 5

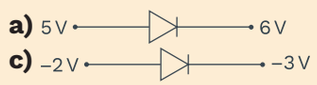
**Q9** A sine wave of peak amplitude 10 V is applied to full wave rectifier and bridge rectifier, respectively. Find peak inverse voltage of FWR and bridge rectifier respectively.

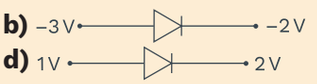
a) 20 V and 10 V b) 10 V and 10 V
c) 20 and 20 V d) 10V and 20 V

**Q10** A sine wave v(t) = 5 sin ωt is applied to centre tap full wave rectifier. Find angle of conduction if practical diode of cut-in voltage 0.6 V is used.

a) 320 b) 327 c) 340 d) 332

**Q11** Which of the following diode is in ON state?

a) 


b) 


c) 


d) 

**Q12** Clipper circuit is used in
a) Wave shaping
b) Clamping
c) Regulator
d) Rectification

**Q13** Clamper circuit is used in
a) Wave shaping
b) Regulator
c) Communication system
d) Rectification

**Q14** The output of voltage multiplier is
a) AC
b) DC
c) AC + DC
d) None of above

**Q15** Zener diode is widely used as
a) Regulator
b) Rectifier
c) Clipper
d) Clamper

**Answer Key:**

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| 1) | a) | 2) | b) | 3) | c) | 4) | c) | 5) | a) |
| 6) | a) | 7) | b) | 8) | c) | 9) | a) | 10) | d) |
| 11) | c) | 12) | a) | 13) | c) | 14) | b) | 15) | a) |